

Digital Notes



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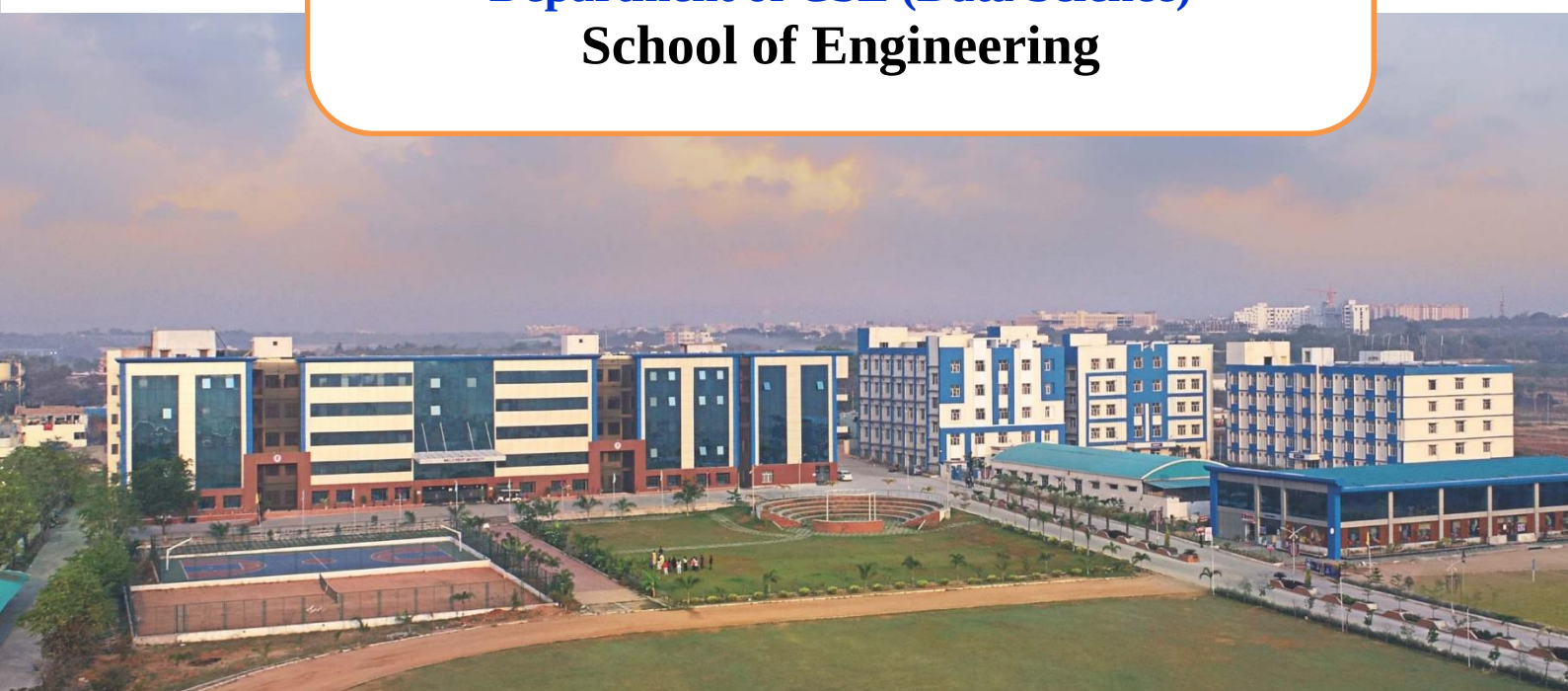
(Telangana State Private Universities Act No.13 of 2020 and
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Data Analytics

**II B.Tech II Semester
(2024-25)**

Department of CSE (Data Science)

School of Engineering



UNIT-1

Introduction to Data Analytics

Overview of Data Analytics and its importance with the Business Domains, Analysis Vs Analytics, Types of Data Analytics, Features and benefits of Data Analytics, Introduction to data types and data sources and its importance, Central Tendency and Dispersion, Applications of Statistics in Data Analytics, Measures of Central Tendency: Mean, Median, Mode, Measures of Dispersion, Skewness and Kurtosis.

What is Data Analytics?

Data Analytics is a systematic approach that transforms raw data into valuable insights. This process encompasses a suite of technologies and tools that facilitate data collection, cleaning, transformation, and modelling, ultimately yielding actionable information. This information serves as a robust support system for decision-making.

Data analysis plays a pivotal role in business growth and performance optimization. It aids in enhancing decision-making processes, bolstering risk management strategies, and enriching customer experiences. By presenting statistical summaries, data analytics provides a concise overview of quantitative data.

While **data analytics** finds extensive application in the finance industry, its utility is not confined to this sector alone. It is also leveraged in diverse fields such as agriculture, banking, retail, and government, among others, underscoring its universal relevance and impact. Thus, data analytics serves as a powerful tool for driving informed decisions and fostering growth across various industries.

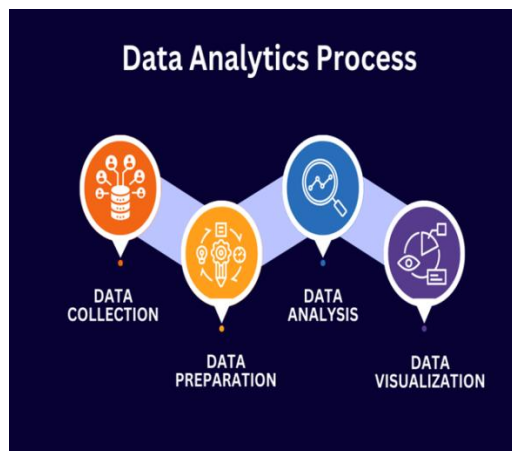
Process of Data Analytics

Data analysts, data scientists, and data engineers together create data pipelines which helps to set up the model and do further analysis. Data Analytics can be done in the following steps which are mentioned below:

1. **Data Collection** : It is the first step where raw data needs to be collected for analysis purposes. It consists of two steps in which [data collection](#) can be done. If the data are from different source systems then using data integration routines the data analysts have to combine the different data

whereas sometimes the data are the subset of the data set. In this case, the data analyst would perform some steps to extract the useful subset and transfer it to the other compartment in the system.

2. **Data Cleansing** : After collecting the data the next step is to clean the quality of the data as the collected data consists of a lot of quality problems such as errors, duplicate entries and white spaces which need to be corrected before moving to the next step. By running data profiling and data cleansing tasks these errors can be corrected. These data are organised according to the needs of the analytical model by the analysts.
3. **Data Analysis and Data Interpretation**: Analytical models are created using software and other tools which interpret the data and understand it. The tools include Python, Excel, R, Scala and [SQL](#). Lastly this model is tested again and again until the model works as it needs to be then in production mode the data set is run against the model.
4. **Data Visualisation**: [Data visualisation](#) is the process of creating visual representation of data using the plots, charts and graphs which helps to analyse the patterns, trends and get the valuable insights of the data. By comparing the datasets and analysing it data analysts find the useful data from the raw data.

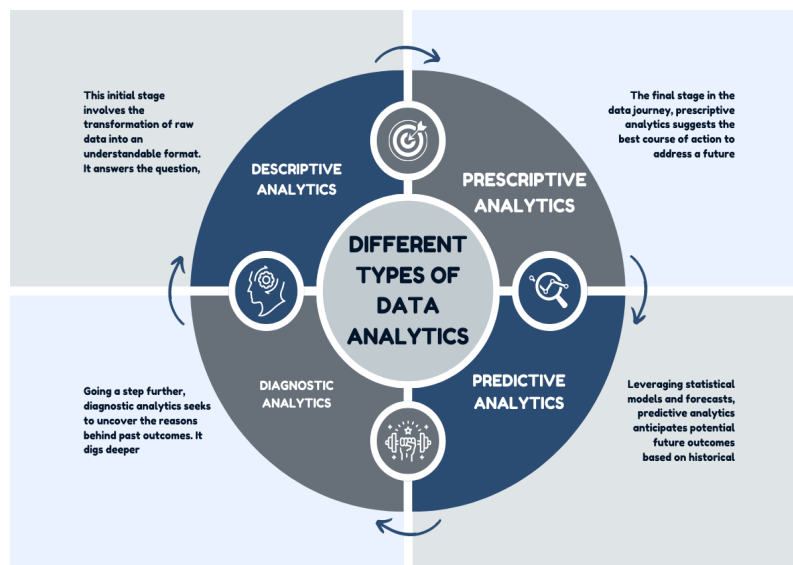


Types of Data Analytics

There are different types of data analysis in which raw data is converted into valuable insights. Some of the types of data analysis are mentioned below:

1. **Descriptive Data Analytics** : Descriptive data Analytics is a type of data analysis which summarises the data set and it is used to compare the past results, differentiate between the weakness and strength, and identify the anomalies. Descriptive data analysis is used by the companies to identify the problems in the data set as it helps in identifying the patterns.

2. **Real-time Data Analytics:** Real time data Analytics doesn't use data from past events. It is a type of data analysis which involves using the data when the data is immediately entered in the database. This type of analysis is used by the companies to identify the trends and track the competitors' operations.
3. **Diagnostic Data Analytics:** Diagnostic Data Analytics uses past data sets to analyse the cause of an anomaly. Some of the techniques used in diagnostic analysis are correlation analysis, regression analysis and analysis of variance. The results which are provided by diagnostic analysis help the companies to give accurate solutions to the problems.
4. **Predictive Data Analytics:** This type of Analytics is done in the current data to predict future outcomes. To build the predictive models it uses machine learning algorithms, statistical model techniques to identify the trends and patterns. Predictive data analysis is also used in sales forecasting, to estimate the risk and to predict customer behaviour.
5. **Prescriptive Data Analytics:** Prescriptive data Analytics is an analysis of selecting best solutions to problems. This type of data analysis is used in loan approval, pricing models, machine repair scheduling, analysing the decisions and so on. To automate decision making companies use prescriptive data analysis.



Methods of Data Analytics

There are two types of methods in data analysis which are mentioned below:

1. Qualitative Data Analytics

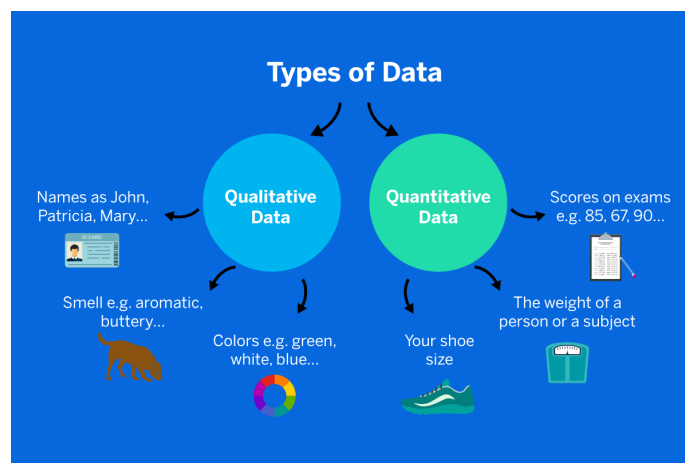
Qualitative data analysis doesn't use statistics and derives data from the words, pictures and symbols. Some common qualitative methods are:

- Narrative Analytics is used for working with data acquired from diaries, interviews and so on.
- Content Analytics is used for Analytics of verbal data and behaviour.
- Grounded theory is used to explain some given event by studying.

2. Quantitative Data Analysis

Quantitative data Analytics is used to collect data and then process it into the numerical data. Some of the quantitative methods are mentioned below:

- Hypothesis testing assesses the given hypothesis of the data set.
- Sample size determination is the method of taking a small sample from a large group of people and then analysing it.
- Average or mean of a subject is dividing the sum total numbers in the list by the number of items present in that list.



Data Analytics and its type

Data analytics is an important field that involves the process of collecting, processing, and interpreting data to uncover insights and help in making decisions. Data analytics is the practice of examining raw data to identify trends, draw conclusions, and extract meaningful information. This involves various techniques and tools to process and transform data into valuable insights that can be used for decision-making.

Understanding Data Analytics

Data analytics encompasses a wide array of techniques for analyzing data to gain valuable insights that can enhance various aspects of operations. By scrutinizing information, businesses can uncover patterns and metrics that

might otherwise go unnoticed, enabling them to optimize processes and improve overall efficiency.

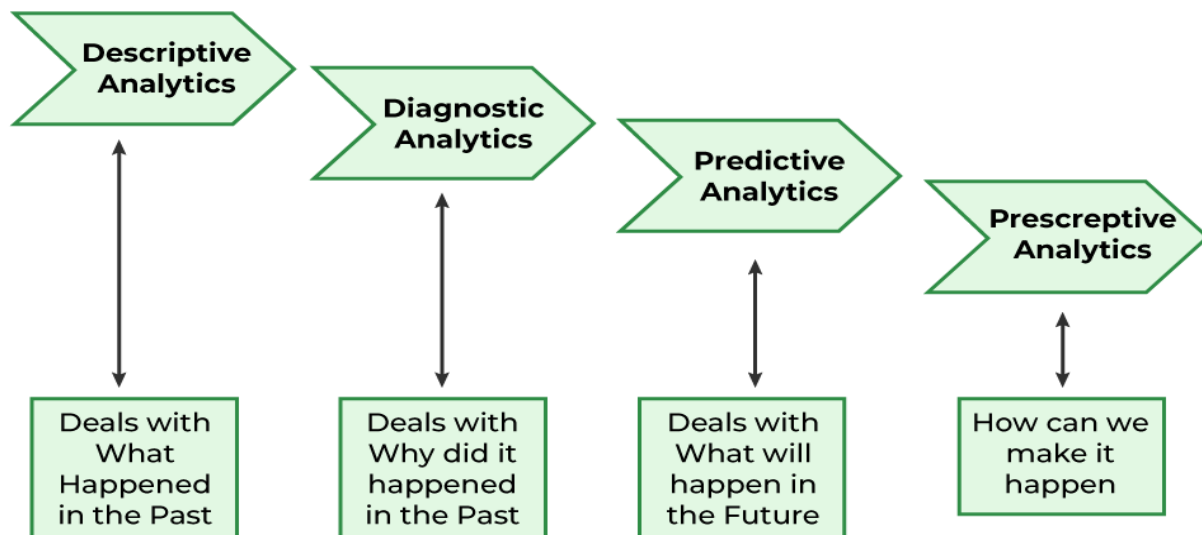
For instance, in manufacturing, companies collect data on machine runtime, downtime, and work queues to analyze and improve workload planning, ensuring machines operate at optimal levels.

Beyond production optimization, data analytics is utilized in diverse sectors. Gaming firms utilize it to design reward systems that engage players effectively, while content providers leverage analytics to optimize content placement and presentation, ultimately driving user engagement.

Types of Data Analytics

There are four major types of data analytics:

1. **Predictive (forecasting)**
2. **Descriptive (business intelligence and data mining)**
3. **Prescriptive (optimization and simulation)**
4. **Diagnostic analytics**



Data Analytics and its Types

Predictive Analytics

Predictive analytics turn the data into valuable, actionable information. predictive analytics uses data to determine the probable outcome of an event or a likelihood of a situation occurring. Predictive analytics holds a variety of statistical techniques from modeling, machine learning, data mining, and game theory that analyze current and historical facts to make predictions about a future event. Techniques that are used for predictive analytics are:

- Linear Regression
- Time Series Analysis and Forecasting
- Data Mining

Basic Cornerstones of Predictive Analytics

- Predictive modeling
- Decision Analysis and optimization
- Transaction profiling

Descriptive Analytics

Descriptive analytics looks at data and analyze past event for insight as to how to approach future events. It looks at past performance and understands the performance by mining historical data to understand the cause of success or failure in the past. Almost all management reporting such as sales, marketing, operations, and finance uses this type of analysis.

The descriptive model quantifies relationships in data in a way that is often used to classify customers or prospects into groups. Unlike a predictive model that focuses on predicting the behavior of a single customer, Descriptive analytics identifies many different relationships between customer and product.

Common examples of Descriptive analytics are company reports that provide historic reviews like:

- Data Queries
- Reports
- Descriptive Statistics
- Data dashboard

Prescriptive Analytics

Prescriptive Analytics automatically synthesize big data, mathematical science, business rule, and machine learning to make a prediction and then suggests a decision option to take advantage of the prediction.

Prescriptive analytics goes beyond predicting future outcomes by also suggesting action benefits from the predictions and showing the decision maker the implication of each decision option. Prescriptive Analytics not only anticipates what will happen and when to happen but also why it will happen. Further, Prescriptive Analytics can suggest decision options on how to take advantage of a future opportunity or mitigate a future risk and illustrate the implication of each decision option.

For example, Prescriptive Analytics can benefit healthcare strategic planning by using analytics to leverage operational and usage data combined with data of external factors such as economic data, population demography, etc.

Diagnostic Analytics

In this analysis, we generally use historical data over other data to answer any question or for the solution of any problem. We try to find any dependency and pattern in the historical data of the particular problem.

For example, companies go for this analysis because it gives a great insight into a problem, and they also keep detailed information about their disposal otherwise data collection may turn out individual for every problem and it will be very time-consuming. Common techniques used for Diagnostic Analytics are:

- Data discovery
- Data mining
- Correlations

The Role of Data Analytics

Data analytics plays a pivotal role in enhancing operations, efficiency, and performance across various industries by uncovering valuable patterns and insights. Implementing data analytics techniques can provide companies with a competitive advantage. The process typically involves four fundamental steps:

- **Data Mining**: This step involves gathering data and information from diverse sources and transforming them into a standardized format for subsequent analysis. Data mining can be a time-intensive process compared to other steps but is crucial for obtaining a comprehensive dataset.
- **Data Management**: Once collected, data needs to be stored, managed, and made accessible. Creating a database is essential for managing the vast amounts of information collected during the mining process. SQL (Structured Query Language) remains a widely used tool for database management, facilitating efficient querying and analysis of relational databases.
- **Statistical Analysis** : In this step, the gathered data is subjected to statistical analysis to identify trends and patterns. Statistical modeling is used to interpret the data and make predictions about future trends. Open-source programming languages like Python, as well as specialized tools like R, are commonly used for statistical analysis and graphical modeling.
- **Data Presentation** : The insights derived from data analytics need to be effectively communicated to stakeholders. This final step involves formatting the results in a manner that is accessible and understandable to various stakeholders, including decision-makers, analysts, and shareholders. Clear and concise data presentation is essential for driving informed decision-making and driving business growth.

Steps in Data Analysis

- **Define Data Requirements** : This involves determining how the data will be grouped or categorized. Data can be segmented based on various factors such as age, demographic, income, or gender, and can consist of numerical values or categorical data.
- **Data Collection** : Data is gathered from different sources, including computers, online platforms, cameras, environmental sensors, or through human personnel.
- **Data Organization** : Once collected, the data needs to be organized in a structured format to facilitate analysis. This could involve using spreadsheets or specialized software designed for managing and analyzing statistical data.
- **Data Cleaning** : Before analysis, the data undergoes a cleaning process to ensure accuracy and reliability. This involves identifying and removing any duplicate or erroneous entries, as well as addressing any missing or incomplete data. Cleaning the data helps to mitigate potential biases and errors that could affect the analysis results.

Usage of Data Analytics

There are some key domains and strategic planning techniques in which Data Analytics has played a vital role:

- **Improved Decision-Making** – If we have supporting data in favour of a decision, then we can implement them with even more success probability. For example, if a certain decision or plan has to lead to better outcomes then there will be no doubt in implementing them again.
 - **Better Customer Service** – Churn modeling is the best example of this in which we try to predict or identify what leads to customer churn and change those things accordingly so, that the attrition of the customers is as low as possible which is a most important factor in any organization.
 - **Efficient Operations** – Data Analytics can help us understand what is the demand of the situation and what should be done to get better results then we will be able to streamline our processes which in turn will lead to efficient operations.
 - **Effective Marketing** – Market segmentation techniques have been implemented to target this important factor only in which we are supposed to find the marketing techniques which will help us increase our sales and leads to effective marketing strategies.
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Key Benefits of Data Analysis

Data Analysis involves inspecting, transforming, and modeling data to discover useful information, inform conclusions, and support decision-making. It encompasses a range of techniques and tools used to interpret raw data, identify patterns, and extract actionable insights. Effective data analysis combines statistical methods, data visualization, and computational techniques to turn data into valuable knowledge.

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The Importance of Data Analysis

Data analysis is pivotal in various aspects of modern life and business. Here's a detailed look at why it is so crucial:

Informed Decision-Making

One of the primary reasons data analysis is important is its role in informed decision-making. By analyzing data, organizations can:

- **Understand Performance:** Analyze sales figures, customer feedback, and operational metrics to gauge performance and make strategic decisions.
- **Predict Outcomes:** Use historical data to forecast future trends, such as sales projections or market demand.
- **Evaluate Strategies:** Assess the effectiveness of marketing campaigns, business strategies, and operational changes.

For example, a retail company might analyze customer purchasing patterns to optimize inventory levels and improve sales strategies.

Improving Business Efficiency

Data analysis helps businesses streamline operations and reduce costs by:

- **Identifying Inefficiencies:** Detect inefficiencies in processes and workflows through data analysis.
- **Optimizing Resources:** Allocate resources more effectively based on data-driven insights.
- **Enhancing Productivity:** Implement process improvements and performance metrics based on data findings.

A manufacturing company, for instance, might use data analysis to optimize its supply chain, leading to reduced costs and improved production efficiency.

Identifying Market Trends

Businesses and organizations use data analysis to stay ahead of market trends:

- **Market Research:** Analyze market data to identify emerging trends, customer preferences, and competitive landscapes.
- **Trend Analysis:** Use data to understand shifts in consumer behavior and adapt strategies accordingly.

A fashion retailer might analyze sales data and social media trends to predict upcoming fashion trends and adjust their product offerings.

Enhancing Customer Experience

Data analysis is crucial for improving customer experiences:

- **Personalization:** Analyze customer data to offer personalized products, services, and recommendations.
- **Feedback Analysis:** Use data from customer feedback and reviews to improve service quality and address issues.

For example, an e-commerce platform might analyze user behavior to offer personalized product recommendations and improve user experience.

Driving Innovation

Data analysis fuels innovation by:

- **Identifying Opportunities:** Spot new business opportunities and areas for improvement.
- **Supporting R&D:** Guide research and development efforts with data-driven insights.

Tech companies often rely on data analysis to drive innovation in product development, such as creating new features based on user feedback and usage patterns.

Supporting Evidence-Based Research

In academic and scientific fields, data analysis supports:

- **Research Validation:** Test hypotheses and validate research findings with statistical methods.
- **Publication of Findings:** Present data-driven evidence in research papers and studies.

Researchers in fields like epidemiology use data analysis to study disease patterns and evaluate public health interventions.

Mitigating Risks

Data analysis helps in identifying and managing risks:

- **Risk Assessment:** Analyze potential risks and vulnerabilities in business operations.
- **Fraud Detection:** Use data analysis techniques to detect fraudulent activities. Financial institutions, for instance, use data analysis to identify suspicious transactions and prevent fraud.

Fostering Competitive Advantage

Data analysis provides a competitive edge by:

- **Benchmarking:** Compare performance against competitors and industry standards.
- **Strategic Planning:** Develop strategic plans based on data-driven insights and competitive analysis.

Businesses that leverage data analysis can gain a competitive advantage by making better strategic decisions and staying ahead of market trends.

Data Analysis VS Data Analytics

1. Data Analytics :

Analytics is a technique of converting raw facts and figures into some particular actions by analyzing those raw data evaluations and perceptions in the context of organizational problem-solving and also with the decision making. Analytics

is the discovery and conversation of significant patterns in data. Especially, precious in areas prosperous with recorded information, analytics depends on the simultaneous utility of statistics, computer programming, and operation lookup to qualify performance. Analytics frequently favors data visualization to talk insight. The aim of Data Analytics is to get actionable insights ensuing in smarter selections and higher commercial enterprise outcomes.

2. Data Analysis :

It is the technique of observing, transforming, cleaning, and modeling raw facts and figures with the purpose of developing beneficial information and acquiring profitable conclusions.

Difference between Data Analytics and Data Analysis:

S.No.	Data Analytics	Data Analysis
1.	It is described as a traditional form or generic form of analytics.	It is described as a particularized form of analytics.
2.	It includes several stages like the collection of data and then the inspection of business data is done.	To process data, firstly raw data is defined in a meaningful manner, then data cleaning and conversion are done to get meaningful information from raw data.
3.	It supports decision making by analyzing enterprise data.	It analyzes the data by focusing on insights into business data.
4.	It uses various tools to process data such as Tableau, Python, Excel, etc.	It uses different tools to analyze data such as Rapid Miner, Open Refine, Node XL, KNIME, etc.
5.	Descriptive analysis cannot be performed on this.	A Descriptive analysis can be performed on this.
6.	One can find anonymous relations with the help of this.	One cannot find anonymous relations with the help of this.
7.	It does not deal with inferential analysis.	It supports inferential analysis

Application of Statistics

Statistics is a branch of mathematics that deals with collecting, analyzing, interpreting, and presenting data. It helps decision-makers to draw evidence-based conclusions from data. There are various application of Statistics mentioned in article below.



Applications of Statistics

- **Healthcare:** In the medical field, statistics are used for designing clinical trials, understanding disease prevalence, and evaluating the effectiveness of treatments. It helps in making informed decisions regarding patient care and public health policies.
- **Business and Economics:** Companies rely on statistical analysis for market research, quality control, and financial forecasting. It aids in understanding consumer behavior, optimizing operations, and assessing economic trends.
- **Environmental Science:** Statistics help in monitoring environmental changes, assessing pollution levels, and studying the impact of human activities on the environment. It's vital for developing sustainable practices and policies.
- **Government and Public Policy:** Statistical data is essential for governments in planning, resource allocation, and policy formulation. It supports decision-making in areas like education, transportation, and social services.
- **Sports:** In sports, statistics are used to analyze player performance, strategize game plans, and predict outcomes. It enhances the understanding of game dynamics and improves team performance.
- **Research and Development:** Statistics are fundamental in scientific research, enabling the testing of hypotheses and interpretation of experimental data. It supports innovation and discovery across all scientific disciplines.

Use of Statistics

Below are some of the uses of Statistics in various domain.

- **Descriptive Statistics:** The company would start by calculating descriptive statistics to understand the overall satisfaction level. They might compute measures like the mean, median, mode, and standard deviation of the ratings. This provides a summary of the central tendency and variability in the data.
- **Data Visualization:** Graphical representations such as histograms, bar charts, or pie charts can visually display the distribution of satisfaction ratings. This helps identify any patterns or trends in customer feedback.
- **Inferential Statistics:** The company may want to infer insights about the entire customer population based on the survey sample. They could use inferential statistics techniques such as hypothesis testing and confidence intervals to make statistical inferences about customer satisfaction levels.
- **Segmentation Analysis:** The company might segment customers based on demographics (age, gender, location) or purchase behavior (frequency, amount spent). Statistical techniques like cluster analysis or ANOVA (Analysis of Variance) can help identify distinct customer segments and their corresponding satisfaction levels.

Applications of Data Analysis

Data analysis has diverse applications across various sectors:

Business and Marketing

- **Customer Segmentation:** Identify different customer groups for targeted marketing strategies.
- **Campaign Effectiveness:** Measure the success of marketing campaigns and adjust tactics as needed.

Healthcare

- **Patient Care:** Analyze patient data for better diagnosis and treatment plans.
- **Public Health:** Study health trends and manage public health initiatives.

Finance

- **Investment Decisions:** Analyze market trends and financial data for investment opportunities.
- **Risk Management:** Assess financial risks and develop strategies for risk mitigation.

Education

- **Student Performance:** Analyze student performance data to improve educational outcomes.

- **Curriculum Development:** Use data to develop and refine educational programs.

Government and Public Policy

- **Policy Analysis:** Evaluate the impact of public policies and programs.
- **Resource Allocation:** Use data to allocate resources effectively in public services.

Challenges in Data Analysis

Despite its importance, data analysis faces several challenges:

Data Quality Issues

- **Incomplete Data:** Missing or incomplete data can lead to inaccurate analysis.
- **Data Accuracy:** Ensuring the accuracy and reliability of data sources.

Data Privacy Concerns

- **Confidentiality:** Protecting sensitive data from unauthorized access and breaches.
- **Compliance:** Adhering to data protection regulations and standards.

Complexity of Data Management

- **Data Integration:** Combining data from various sources can be complex and time-consuming.
- **Data Storage:** Managing large volumes of data and ensuring efficient storage solutions.

Skills and Expertise Requirements

- **Technical Skills:** Requires expertise in statistical analysis, data management, and programming.
- **Interpretation Skills:** Analyzing data is not enough; interpreting results accurately is crucial.

Future Trends in Data Analysis

The field of data analysis is evolving with several emerging trends:

Artificial Intelligence and Machine Learning

- **Advanced Analytics:** AI and ML techniques are used for more sophisticated data analysis and predictive modeling.
- **Automation:** Automating repetitive tasks and processes through AI-driven tools.

Big Data Analytics

- **Scalability:** Handling and analyzing vast amounts of data from diverse sources.
- **Advanced Techniques:** Utilizing advanced algorithms for big data processing and analysis.

Real-Time Data Processing

- **Immediate Insights:** Analyzing data in real-time for timely decision-making.

- **Applications:** Real-time data analysis is used in finance, transportation, and social media.

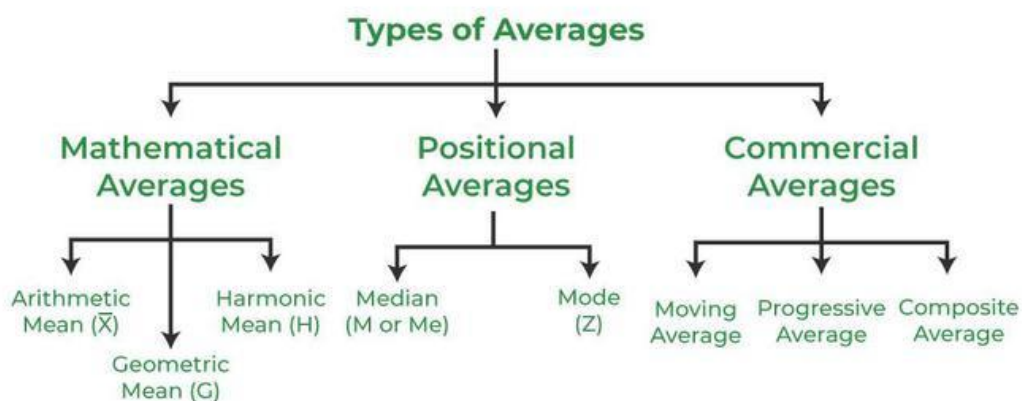
Increased Emphasis on Data Ethics

- **Ethical Practices:** Ensuring ethical practices in data collection, analysis, and usage.
- **Transparency:** Promoting transparency and accountability in data analysis processes.

Measures of Central Tendency and Dispersion

Usually, frequency distribution and graphical representation are used to depict a set of raw data to attain meaningful conclusions from them. However, sometimes, these methods fail to convey a proper and clear picture of the data as expected. Therefore, some measures, also known as **Measures of Central Tendency** or **Average** are used as a single measurement to determine the main characteristics of the given series. Hence, the **Measure of Central Tendency** is a single value used for the representation of a complete set of data. Another name for Measure of Central Tendency is **Measure of Location**. Average is a typical value of the given data set to which most of the observations of the data fall closer than any other value. The three principal measures that are used in Statistical Analysis are **Arithmetic Mean, Median, and Mode**.

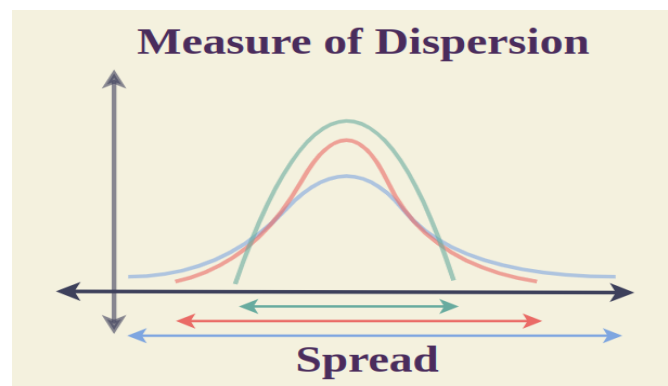
The different measures of central tendency can be classified into three categories; viz., Mathematical Averages (Arithmetic Mean($\bar{X}$), Geometric Mean(G), and Harmonic Mean(H)), Positional Averages (Median(M or Me) and Mode(Z)), and Commercial Averages (Moving Average, Progressive Average, and Composite Average).



Measures of central tendency and dispersion are statistical measures used to describe the characteristics of a dataset. Central tendency helps us identify a single representative value around which data tends to cluster, whereas measures of dispersion tell us how deviated data is from the central value. Measures of central tendency include mean, median, and mode, while measures of dispersion include range, variance, standard deviation, and interquartile range.

Measures of Dispersion | Types, Formula and Examples

- **Measures of Dispersion** are used to represent the scattering of data. These are the numbers that show the various aspects of the data spread across various parameters.



Dispersion in Statistics

Dispersion in statistics is a way to describe how spread out or scattered the data is around an average value. It helps to understand if the data points are close together or far apart.

Dispersion shows the variability or consistency in a set of data. There are different measures of dispersion like range, variance, and standard deviation.

Measure of Dispersion in Statistics

Measures of Dispersion measure the scattering of the data. It tells us how the values are distributed in the data set. In statistics, we define the measure of dispersion as various parameters that are used to define the various attributes of the data.

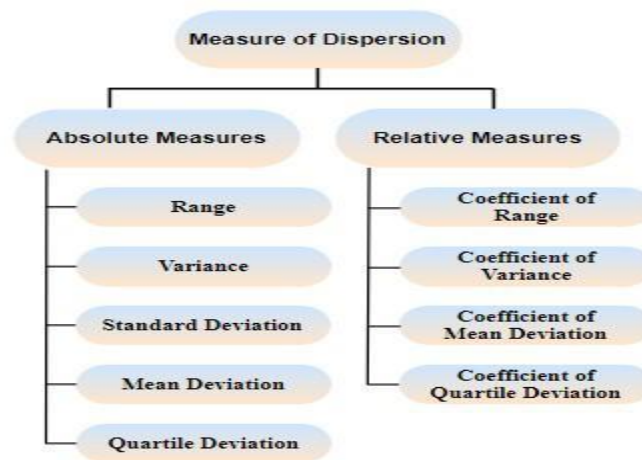
These measures of dispersion capture variation between different values of the data.

Types of Measures of Dispersion

Measures of dispersion can be classified into the following two types :

- Absolute Measure of Dispersion
- Relative Measure of Dispersion

These measures of dispersion can be further divided into various categories. They have various parameters and these parameters have the same unit.



Absolute Measure of Dispersion

The measures of dispersion that are measured and expressed in the units of data themselves are called Absolute Measure of Dispersion. For example – Meters, Dollars, Kg, etc.

Some absolute measures of dispersion are:

Range: It is defined as the difference between the largest and the smallest value in the distribution.

Mean Deviation: It is the arithmetic mean of the difference between the values and their mean.

Standard Deviation: It is the square root of the arithmetic average of the square of the deviations measured from the mean.

Variance: It is defined as the average of the square deviation from the mean of the given data set.

Quartile Deviation: It is defined as half of the difference between the third quartile and the first quartile in a given data set.

Interquartile Range: The difference between upper(Q_3) and lower(Q_1) quartile is called Interquartile Range. Its formula is given as $Q_3 - Q_1$.

Relative Measure of Dispersion

We use relative measures of dispersion to measure the two quantities that have different units to get a better idea about the scattering of the data.

Here are some of the relative measures of dispersion:

Coefficient of Range: It is defined as the ratio of the difference between the highest and lowest value in a data set to the sum of the highest and lowest value.

Coefficient of Variation: It is defined as the ratio of the standard deviation to the mean of the data set. We use percentages to express the coefficient of variation.

Coefficient of Mean Deviation: It is defined as the ratio of the mean deviation to the value of the central point of the data set.

Coefficient of Quartile Deviation: It is defined as the ratio of the difference between the third quartile and the first quartile to the sum of the third and first quartiles.

Range of Data Set

The range is the difference between the largest and the smallest values in the distribution.

Thus, it can be written as

$$R = L - S$$

where,

L is the largest value in the Distribution

S is the smallest value in the Distribution

- A higher value of range implies higher variation in the data set.
- One drawback of this measure is that it only takes into account the maximum and the minimum value. They might not always be the proper indicator of how the values of the distribution are scattered.

Example: Find the range of the data set 10, 20, 15, 0, 100.

Solution:

- *Smallest Value in the data = 0*
- *Largest Value in the data = 100*

Thus, the range of the data set is,

$$R = 100 - 0$$

$$R = 100$$

Note: Range cannot be calculated for the open-ended frequency distributions. Open-ended frequency distributions are those distributions in which either the

lower limit of the lowest class or the higher limit of the highest class is not defined.

Range for Ungrouped Data

To find the range for the ungrouped data set, first we have to find the smallest and the largest value of the data set by observing. The difference between them gives the range of ungrouped data.

We can understand this with the help of following example:

Example: Find out the range for the following observations, 20, 24, 31, 17, 45, 39, 51, 61.

Solution:

- *Largest Value = 61*
- *Smallest Value = 17*

Thus, the range of the data set is

$$\text{Range} = 61 - 17 = 44$$

Range for Grouped Data

The range of the grouped data set is found by studying the following example,

Example: Find out the range for the following frequency distribution table for the marks scored by class 10 students.

Marks Intervals	Number of Students
0-10	5
10-20	8
20-30	15
30-40	9

Solution:

- *For Largest Value: Taking the higher limit of Highest Class = 40*
- *For Smallest Value: Taking the lower limit of Lowest Class = 0*

$$\text{Range} = 40 - 0$$

Thus, the range of the given data set is,
 $Range = 40$

Mean Deviation

Mean deviation measures the deviation of the observations from the mean of the distribution.

Since the average is the central value of the data, some deviations might be positive and some might be negative. If they are added like that, their sum will not reveal much as they tend to cancel each other's effect.

For example :

Let us consider this set of data : -5, 10, 25

Mean = $(-5 + 10 + 25)/3 = 10$

Now a deviation from the mean for different values is,

- $(-5 - 10) = -15$
- $(10 - 10) = 0$
- $(25 - 10) = 15$

Now adding the deviations, shows that there is zero deviation from the mean which is incorrect. Thus, to counter this problem only the absolute values of the difference are taken while calculating the mean deviation.

Mean Deviation Formula :

$$MD = \frac{1}{n} \left[\frac{(x_1 - \mu)}{n} + \frac{(x_2 - \mu)}{n} + \dots + \frac{(x_n - \mu)}{n} \right]$$

Mean Deviation for Ungrouped Data

For calculating the mean deviation for ungrouped data, the following steps must be followed:

Step 1: Calculate the arithmetic mean for all the values of the dataset.

Step 2: Calculate the difference between each value of the dataset and the mean. Only absolute values of the differences will be considered. $|d|$

Step 3: Calculate the arithmetic mean of these deviations using the formula,

$$M.D = \frac{\sum |d|}{n}$$

This can be explained using the example.

Example: Calculate the mean deviation for the given ungrouped data, 2, 4, 6, 8, 10

Solution:

$$Mean(\mu) = (2+4+6+8+10)/(5)$$

$$\mu = 6$$

$$M. D = \frac{\sum |d|}{n}$$

$$\Rightarrow M.D = \frac{1}{5} [(2-6) + (4-6) + (6-6) + (8-6) + (10-6)] = \frac{5}{5} = 1$$

$$\Rightarrow M.D = \frac{4+2+0+2+4}{5}$$

$$\Rightarrow M.D = \frac{12}{5} = 2.4$$

Measures of Dispersion Formula

Measures of Dispersion Formulas are used to tell us about the various parameters of the data. Various formulas related to the measures of dispersion are discussed in the table below.

Absolute Measures of Dispersion	Related Formulas
Formulae of Measures of Dispersion	
Range	H – S where, H is the Largest Value and S is the Smallest Value
Variance	Population Variance, $\sigma^2 = \frac{\sum (x_i - \mu)^2}{n}$ Sample Variance, $S^2 = \frac{\sum (x_i - \mu)^2}{(n-1)}$ where, μ is the mean and n is the number of observation
Standard Deviation	S.D. = $\sqrt{\sigma^2}$
Mean Deviation	$\mu = \frac{\sum (x - a)}{n}$ where, a is the central value(mean, median, mode) and n is the number of observation
Quartile Deviation	$\frac{(Q_3 - Q_1)}{2}$ where, Q3 = Third Quartile and Q1 = First Quartile

Coefficient of Dispersion

Coefficients of dispersion are calculated when two series are compared, which have great differences in their average. We also use co-efficient of dispersion for comparing two series that have different measurements. It is denoted using the letters C.D.

Relative Measures of Dispersion	Related Formulas
Coefficient of Range	$(H - S)/(H + S)$
Coefficient of Variation	$(SD/\text{Mean}) \times 100$
Coefficient of Mean Deviation	$(\text{Mean Deviation})/\mu$ where, μ is the central point for which the mean is calculated
Coefficient of Quartile Deviation	$(Q3 - Q1)/(Q3 + Q1)$

Measures of Dispersion and Central Tendency

Both Measures of Dispersion and Central Tendency are numbers that are used to describe various parameters of the data. Let's see the differences between Measures of Dispersion and Central Tendency.

Central Tendency	Measure of Dispersion
Central Tendency vs. Measure of Dispersion	
Central Tendency is a term used for the numbers that quantify the properties of the data set.	Measure of Distribution is used to quantify the variability of the data of dispersion.

Central Tendency	Measure of Dispersion
Measure of Central tendency include, • Mean • Median • Mode	Various parameters included for the measure of dispersion are, • Range • Variance • Standard Deviation • Mean Deviation • Quartile Deviation

What is Central Tendency?

Central tendency is a statistical concept that refers to the tendency of data to cluster around a central value or a typical value.

It identifies a single value as representative of an entire data distribution. In other words, the [central tendency](#) is a way of describing the centre or midpoint of a dataset. The three most common measures of central tendency are:

- [Mean](#)
- [Median](#)
- [Mode](#)

Mean

The mean, also known as the average, is calculated by summing up all the values in a dataset and dividing the sum by the total number of values. It is sensitive to extreme values, making it susceptible to outliers.

Median

The median is the middle value in a dataset when the values are arranged in ascending or descending order. If there is an even number of values, the median is the average of the two middle values. Unlike the mean, the median is less affected by outliers.

Mode

The mode is the value that occurs most frequently in a dataset. It is particularly useful for categorical data but can also be applied to numerical data. A dataset may have one mode (unimodal), two modes (bimodal), or more than two modes (multimodal).

Note: The choice of which central tendency measure to use depends on the properties of the data. For example, the mean is best for symmetric distributions, while the median is better for skewed distributions with outliers. The mode is useful for categorical data.

Example:

Let's consider a dataset of daily temperatures recorded over a week: 22°C, 23°C, 21°C, 25°C, 22°C, 24°C, and 20°C.

- **Mean:** $(22 + 23 + 21 + 25 + 22 + 24 + 20) / 7 = 21.86^\circ\text{C}$
- **Median:** Arranging the temperatures in ascending order: 20°C, 21°C, 22°C, 22°C, 23°C, 24°C, 25°C. The median is 22°C.
- **Mode:** The mode is 22°C as it occurs most frequently in the dataset.

What is Dispersion?

Dispersion, also known as variability or spread, measures the extent to which individual data points deviate from the central value. It provides information about the spread or distribution of data points in a dataset.

Common measures of dispersion include

- [Range](#)
- [Variance](#)
- [Standard Deviation](#)
- [Interquartile Range \(IQR\)](#)

Range

In statistics, the range refers to the difference between the highest and lowest values in a dataset. It provides a simple measure of variability, indicating the spread of data points. The range is calculated by subtracting the lowest value from the highest value.

For example, in a dataset {4, 6, 9, 3, 7}, the range is $9 - 3 = 6$.

Variance

Variance is a statistical measure that quantifies the amount of variation or dispersion of a set of numbers from their mean value. Specifically, variance is defined as the expected value of the squared deviation from the mean. It is calculated by:

1. Finding the mean (average) of the data set.
2. Subtracting the mean from each data point to get the deviations from the mean.
3. Squaring each of the deviations.
4. Calculating the average of the squared deviations. This is the variance.

Standard Deviation

Standard deviation is a measure of the amount of variation or dispersion of a set of values from the mean value. It is calculated as the square root of the variance, which is the average squared deviation from the mean.

Interquartile Range (IQR)

Interquartile Range (IQR) is a measure of statistical dispersion that represents the middle 50% of a data set. It is calculated as the difference between the 75th percentile (Q₃) and the 25th percentile (Q₁) of the data i.e., $IQR = Q_3 - Q_1$.

Examples for Dispersion

Let's consider the same dataset of daily temperatures recorded over a week: 22°C, 23°C, 21°C, 25°C, 22°C, 24°C, and 20°C.

Range: Maximum temperature - Minimum temperature = 25°C - 20°C = 5°C

Variance: Variance = (Sum of squared differences from the mean) / (Number of data points)

Mean = 21.86 °C

Sum of squared differences from the mean = $(22 - 21.86)^2 + (23 - 21.86)^2 + (21 - 21.86)^2 + (25 - 21.86)^2 + (22 - 21.86)^2 + (24 - 21.86)^2 + (20 - 21.86)^2$
= $(0.14)^2 + (1.14)^2 + (-0.86)^2 + (3.14)^2 + (0.14)^2 + (2.14)^2 + (-1.86)^2$
= $0.0196 + 1.2996 + 0.7396 + 9.8596 + 0.0196 + 4.5796 + 3.4596$
= 19.0772

Thus, Variance = $19.0772 / 7 \approx 2.725$ °C

Standard Deviation: Take the square root of the variance to get the standard deviation.

Thus, Standard Deviation $\approx \sqrt{2.725} \approx 1.65$ °C

Interquartile Range (IQR): First Quartile (Q₁) = 21°C Third Quartile (Q₃) = 24°C

Thus, $IQR = Q_3 - Q_1 = 24^\circ\text{C} - 21^\circ\text{C} = 3^\circ\text{C}$

Differences between Central Tendency and Dispersion

The key differences between central tendency and dispersion are listed in the following table:

Aspect	Central Tendency	Dispersion
Definition	Indicates the typical or central value around which data tends to cluster	Indicates the spread or variability of data points in a dataset
Purpose	Provides a single representative value summarizing the dataset	Describes how spread out the values are from each other and from the central value
Examples	Mean, median, mode	Range, variance, standard deviation, interquartile range
Calculation	Calculated using the values of the dataset	Calculated using deviations from the central value (usually the mean)
Interpretation	Helps in understanding the center of the data distribution	Helps in understanding the spread of the data distribution
Measure of Location	Indicates where the data is centered	Indicates how widely the data is spread out

UNIT-II

Data Exploration and Visualization: Data Preparation & preprocessing, exploratory data analysis (EDA) techniques, Data visualization using graphs, charts.

Sampling Methods: Population vs. sample, Sampling Types & Techniques, Probability Random sampling vs Non-Probability Random Sampling, Probability sampling methods: Simple random sampling, Cluster sampling, Stratified, Systematic. Non-probability sampling methods: Convenience sampling, Snowball sampling, Quota sampling, Voluntary response sampling.

Introduction to Data Preparation

- Data preparation is the process of making raw data ready for after processing and analysis.
- The key methods are to collect, clean, and label raw data in a format suitable for machine learning (ML) algorithms, followed by data exploration and visualization.
- The process of cleaning and combining raw data before using it for Machine Learning and business analysis is known as data preparation, or sometimes "pre-processing."
- But it may not be the most attractive of duties, careful data preparation is essential to the success of data analytics. Clear and important ideas from raw data require careful validation, cleaning, and an addition.
- Any business analysis or model created will only be as strong and validating as the very first information preparation.

Data Preprocessing

- Data preprocessing is a basic and primary step for converting raw data into useful information.
- It refers to the cleaning, transforming, and integrating of data in order to make it ready for analysis.
- Data preprocessing transforms the data into a format that is more easily and effectively processed in data mining, machine learning and other data science tasks.

Importance of Data Preparation

Data preparation acts as the foundation for successful machine learning projects as: Improves Data Quality: Raw data often contains inconsistencies, missing values, errors, and irrelevant information.

Data preparation techniques like cleaning, imputation, and normalization address these issues, resulting in a cleaner and more consistent dataset. This, in turn, prevents these issues from biasing or hindering the learning process of your models.

Enhances Model Performance: Machine learning algorithms rely heavily on the quality of the data they are trained on. By preparing your data effectively, you provide the algorithms with a clear and well-structured foundation for learning patterns and relationships. This leads to models that are better able to generalize and make accurate predictions on unseen data.

Saves Time and Resources: Investing time upfront in data preparation can significantly save time and resources down the line. By addressing data quality issues early on, you avoid encountering problems later in the modelling process that might require re-work or troubleshooting. This translates to a more efficient and streamlined machine learning workflow

Facilitates Feature Engineering: Data preparation often involves feature engineering, which is the process of creating new features from existing ones. These new features can be more informative and relevant to the task at hand, ultimately improving the model's ability to learn and make predictions.

Data

Preprocessing

It ensures data is accurate, consistent, and reliable

It makes data easier to understand for machines

It helps to ensure data is suitable for business intelligence, analytics, and decision-making.



Key aspects of data preprocessing include:

Data Cleaning: Handling missing values, removing duplicates, and correcting errors.

Data Transformation: Scaling, normalization, encoding categorical variables, and feature engineering.

Data Reduction: Selecting relevant features and reducing dimensionality.

Data Integration: Combining data from multiple sources.

Data Formatting: Ensuring consistent data types and structures.

Steps Involved in Data Preprocessing

a) Data Collection and Import

The first step in any data preprocessing pipeline is gathering the necessary data. This may involve querying databases, accessing APIs, scraping websites, or importing data from various file formats like CSV, JSON, or Excel.

b) Data Exploration and Profiling

Before getting into cleaning and transformation, it's essential to understand your data. This involves examining the structure of your dataset, checking data types, looking for patterns, and identifying potential issues.

c) Data Cleaning

This step involves handling missing data, removing duplicates, correcting errors, and dealing with outliers.

Handling missing data: You might choose to drop rows with missing values, fill them with a specific value (like the mean or median), or use more advanced imputation techniques.

Removing duplicates: Duplicate records can skew your analysis and should be removed.

Correcting errors: This might involve fixing typos, standardizing formats (e.g., date formats), or correcting impossible values.

Dealing with outliers: Outliers can be legitimate extreme values or errors. You need to investigate them and decide whether to keep, modify, or remove them.

d) Data Transformation

This step involves modifying the data to make it more suitable for analysis or modeling. Common transformations include:

Normalization or standardization: Scaling numerical features to a common range.

Encoding categorical variables: Converting categorical data into numerical format.

Feature engineering: Creating new features from existing ones.

Handling skewed data: Applying transformations like log or square root to make the distribution more normal.

e) Data Reduction

For large datasets, it might be necessary to reduce the volume of data while preserving as much information as possible. This can involve:

Feature selection: Choosing the most relevant features for your analysis.

Dimensionality reduction: Using techniques like Principal Component Analysis

(PCA) to reduce the number of features.

Sampling: Working with a representative subset of your data.

f) **Data Validation**

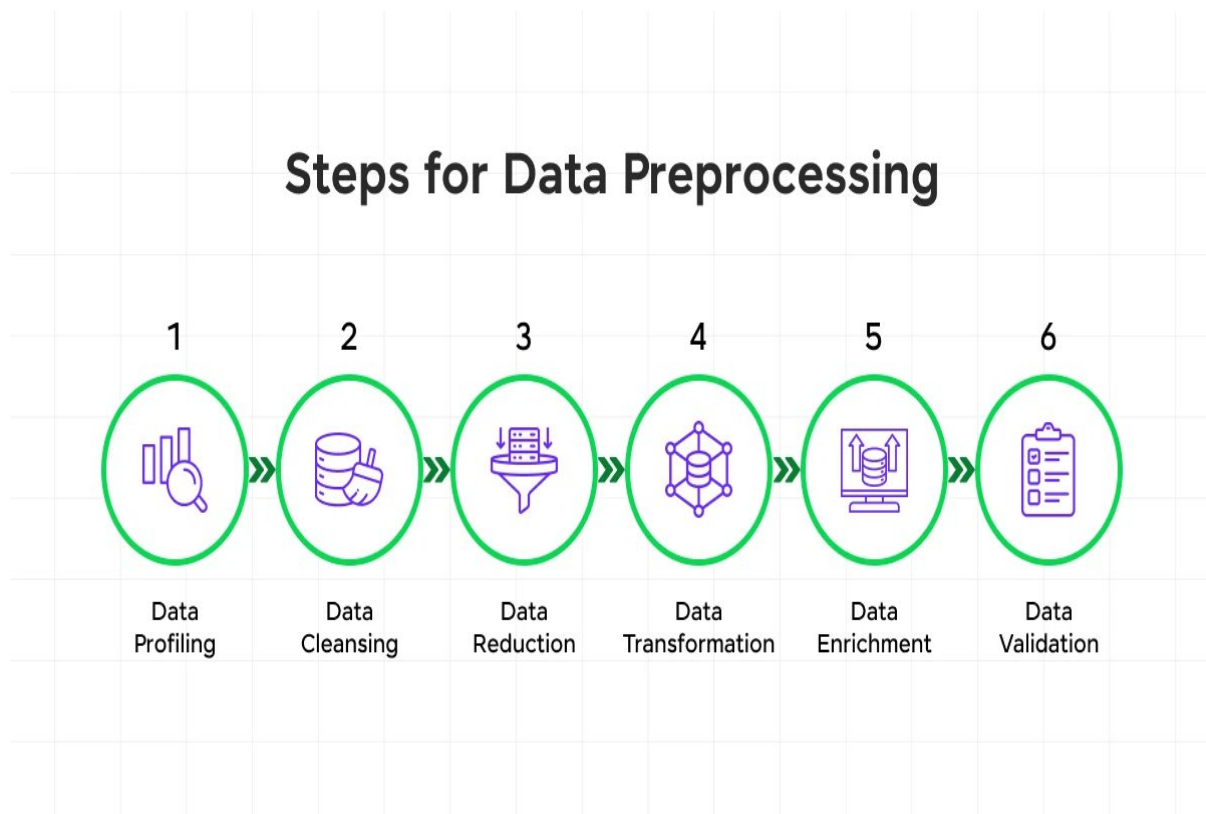
The final step is to validate your preprocessed data to ensure it meets the requirements for your analysis or modeling task. This might involve:

Checking data types

Verifying value ranges

Ensuring all necessary features are present

Checking for any remaining missing values or inconsistencies



Visual Representation of the Data

Data visualization translates complex data sets into visual formats that are easier for the human brain to comprehend. This can include a variety of visual tools such as:

- Charts: Bar charts, line charts, pie charts, etc.
- Graphs: Scatter plots, histograms, etc.
- Maps: Geographic maps, heat maps, etc.
- Dashboards: Interactive platforms that combine multiple visualizations.

The primary goal of data visualization is to make data more accessible and easier to interpret, allowing users to identify patterns, trends, and outliers quickly. This is particularly important

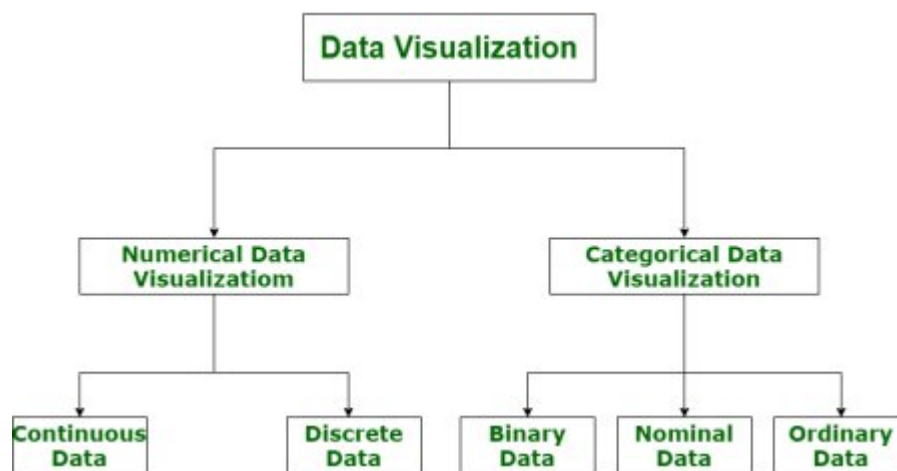
in the context of big data, where the sheer volume of information can be overwhelming without effective visualization techniques.

Types of Data for Visualization

Performing accurate visualization of data is very critical to market research where both numerical and categorical data can be visualized, which helps increase the impact of insights and also helps in reducing the risk of analysis paralysis. So, data visualization is categorized into the following categories:

- Numerical Data
- Categorical Data

Let's understand the visualization of data via a diagram with its all categories.



visual data representations

- Histogram -- vertical bar chart of frequencies
- Frequency Polygon -- line graph of frequencies
- Ogive -- line graph of cumulative frequencies
- Pie Chart -- proportional representation for categories of a whole
- Stem and Leaf Plot
- Pareto Chart
- Scatter Plot

Chart Type	Description	Best Used For	Visual Characteristics
Histogram	Vertical bars showing frequency of data in intervals	Distribution of continuous data	Bars touch each other
Frequency Polygon	Line graph connecting midpoints of histogram bars	Comparing distributions	Line connecting points over histogram bars
Ogive	Line graph of cumulative frequencies	Percentiles, medians, cumulative analysis	Rising line graph
Pie Chart	Circular chart showing proportions of categories	Categorical data, parts of a whole	Circular “slices” representing percentages
Stem and Leaf Plot	Split digits showing both frequency and data values	Small datasets, preserving raw values	Numbers split into "stem" and "leaf"
Pareto Chart	Descending bar chart with cumulative line	Prioritizing issues (80/20 rule)	Tallest-to-shortest bars + cumulative line
Scatter Plot	Plot of paired data points	Relationships/correlations between two variables	Dots in a Cartesian plane, often showing patterns

Why is Data Visualization Important?

1. Data Visualization Discovers the Trends in Data

The most important thing that data visualization does is discover the trends in data. After all, it is much easier to observe data trends when all the data is laid out in front of you in a visual form as compared to data in a table. For example, the screenshot below on visualization on Tableau demonstrates the sum of sales made by each customer in descending order. However, the color red denotes loss while grey denotes profits. So it is very easy to observe from this visualization that even though some customers may have huge sales, they are still at a loss. This would be very difficult to observe from a table.

2. Data Visualization Provides a Perspective on the Data

Visualizing Data provides a perspective on data by showing its meaning in the larger scheme of things. It demonstrates how particular data references stand concerning the overall data picture. In the data visualization below, the data between sales and profit provides a data perspective concerning these two measures. It also demonstrates that there are very few sales above 12K and higher sales do not necessarily mean a higher profit.

3. Data Visualization Puts the Data into the Correct Context

It isn't easy to understand the context of the data with data visualization. Since context provides the whole circumstances of the data, it is very difficult to grasp by just reading numbers in a table. In the below data visualization on Tableau, a TreeMap is used to

demonstrate the number of sales in each region of the United States. It is very easy to understand from this data visualization that California has the largest number of sales out of the total number since the rectangle for California is the largest. But this information is not easy to understand outside of context without visualizing data.

4. Data Visualization Saves Time

It is definitely faster to gather some insights from the data using data visualization rather than just studying a chart. In the screenshot below on Tableau, it is very easy to identify the states that have suffered a net loss rather than a profit. This is because all the cells with a loss are coloured red using a heat map, so it is obvious states have suffered a loss. Compare this to a normal table where you would need to check each cell to see if it has a negative value to determine a loss. Visualizing Data can save a lot of time in this situation.

5. Data Visualization Tells a Data Story

Data visualization is also a medium to tell a data story to the viewers. The visualization can be used to present the data facts in an easy-to-understand form while telling a story and leading the viewers to an inevitable conclusion. This data story, like any other type of story, should have a good beginning, a basic plot, and an ending that it is leading towards. For example, if a data analyst has to craft a data visualization for company executives detailing the profits of various products, then the data story can start with the profits and losses of multiple products and move on to recommendations on how to tackle the losses.

Tools for Visualization of Data

The following are the 10 best Data Visualization Tools

1. Tableau
2. Looker
3. Zoho Analytics
4. Sisense
5. IBM Cognos Analytics
6. Qlik Sense
7. Domo
8. Microsoft Power BI
9. Klipfolio
10. SAP Analytics Cloud

Advantages and Disadvantages of Data Visualization

Advantages of Data Visualization:

- Enhanced Comparison: Visualizing performances of two elements or scenarios streamlines analysis, saving time compared to traditional data examination.
- Improved Methodology: Representing data graphically offers a superior understanding of situations, exemplified by tools like Google Trends illustrating industry trends in graphical forms.
- Efficient Data Sharing: Visual data presentation facilitates effective communication, making information more digestible and engaging compared to sharing raw data.

- Sales Analysis: Data visualization aids sales professionals in comprehending product sales trends, identifying influencing factors through tools like heat maps, and understanding customer types, geography impacts, and repeat customer behaviors.
- Identifying Event Relations: Discovering correlations between events helps businesses understand external factors affecting their performance, such as online sales surges during festive seasons.
- Exploring Opportunities and Trends: Data visualization empowers business leaders to uncover patterns and opportunities within vast datasets, enabling a deeper understanding of customer behaviors and insights into emerging business trends.\

Disadvantages of Data Visualization:

- Can be time-consuming: Creating visualizations can be a time-consuming process, especially when dealing with large and complex datasets.
- Can be misleading: While data visualization can help identify patterns and relationships in data, it can also be misleading if not done correctly. Visualizations can create the impression of patterns or trends that may not exist, leading to incorrect conclusions and poor decision-making.
- Can be difficult to interpret: Some types of visualizations, such as those that involve 3D or interactive elements, can be difficult to interpret and understand.
- May not be suitable for all types of data: Certain types of data, such as text or audio data, may not lend themselves well to visualization. In these cases, alternative methods of analysis may be more appropriate.
- May not be accessible to all users: Some users may have visual impairments or other disabilities that make it difficult or impossible for them to interpret visualizations. In these cases, alternative methods of presenting data may be necessary to ensure accessibility.

SAMPLING METHODS

Population and Sample

The population gives a complete picture, while the sample provides an estimate. Parameters (like population mean) describe the population; statistics (like sample mean) describe the sample. The population refers to the entire group of individuals or items that we are interested in studying and drawing conclusions about. In statistics, the population is the entire set of items from which data is drawn in the statistical study. It can be a group of individuals or a set of items. The population is usually denoted by N .

A sample is a subset of the population selected for study. It is a representative portion of the population from which we collect data in order to make inferences or draw conclusions about the entire population. The sample is denoted by n .

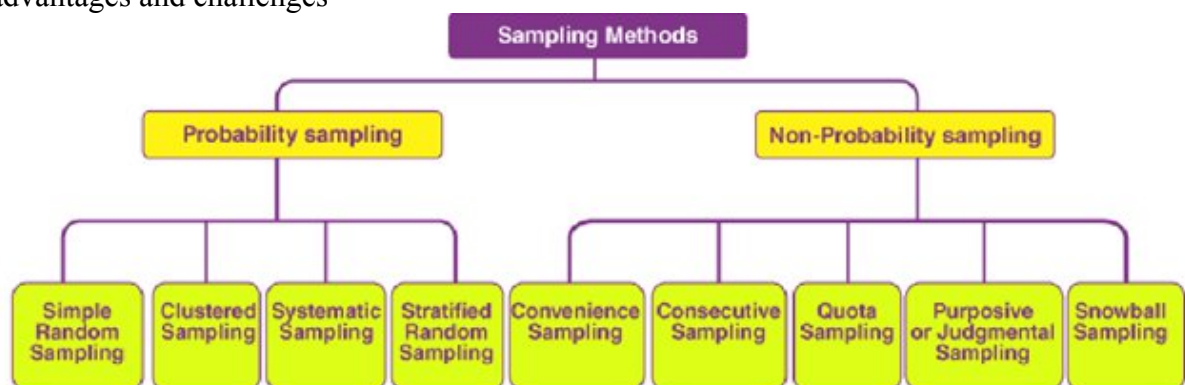
Population	Sample
The population includes all members of a specified group.	A sample is a subset of the population.
Collecting data from an entire population can be time-consuming, expensive, and sometimes impractical or impossible.	Samples offer a more feasible approach to studying populations, allowing researchers to draw conclusions based on smaller, manageable datasets
Includes all residents in the city.	Consists of 1000 households, a subset of the entire population.

Sampling Types & Techniques

Data sampling is a statistical method that involves selecting a part of a population of data to create representative samples. The fundamental aim is to draw conclusions about the entire population without having to engage with every individual data point, thus saving time, resources, and effort while still achieving accurate results.

Types of Data Sampling Methods

Sampling techniques are categorized into two main types: probability sampling and non-probability sampling. Each type is tailored to specific research needs and offers unique advantages and challenges.



Probability Sampling

- Simple Random Sampling
- Stratified Sampling
- Cluster Sampling
- Systematic Sampling

Non-Probability Sampling

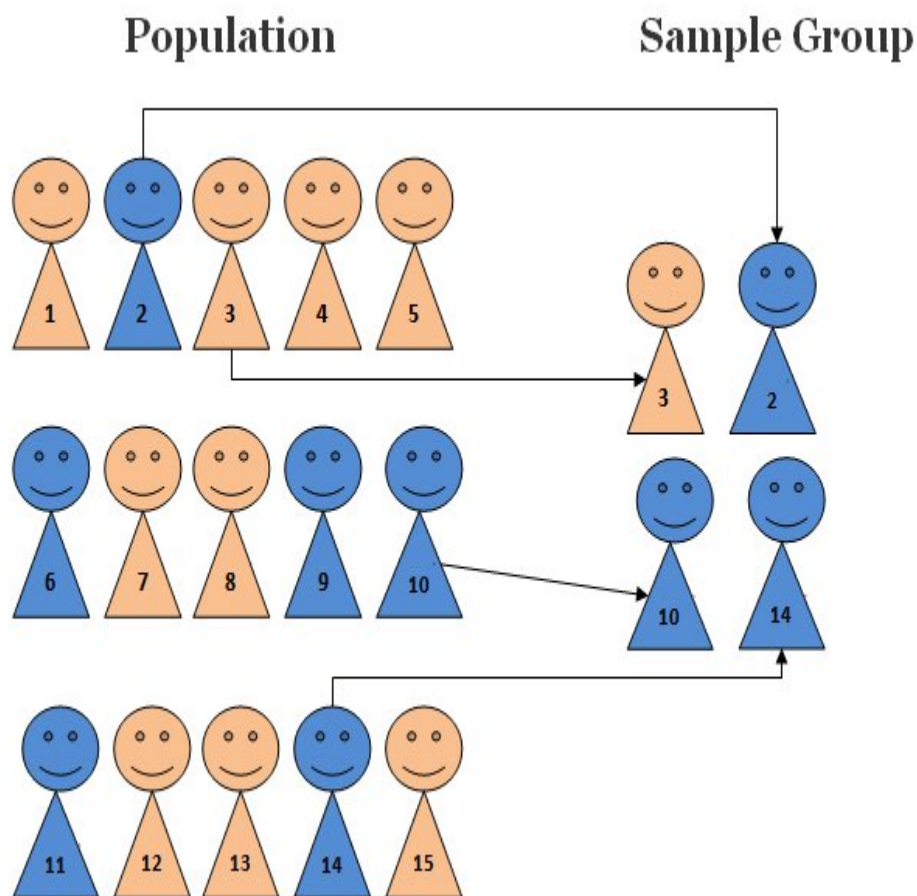
- Convenience Sampling
- Purposive Sampling
- Snowball Sampling
- Quota Sampling

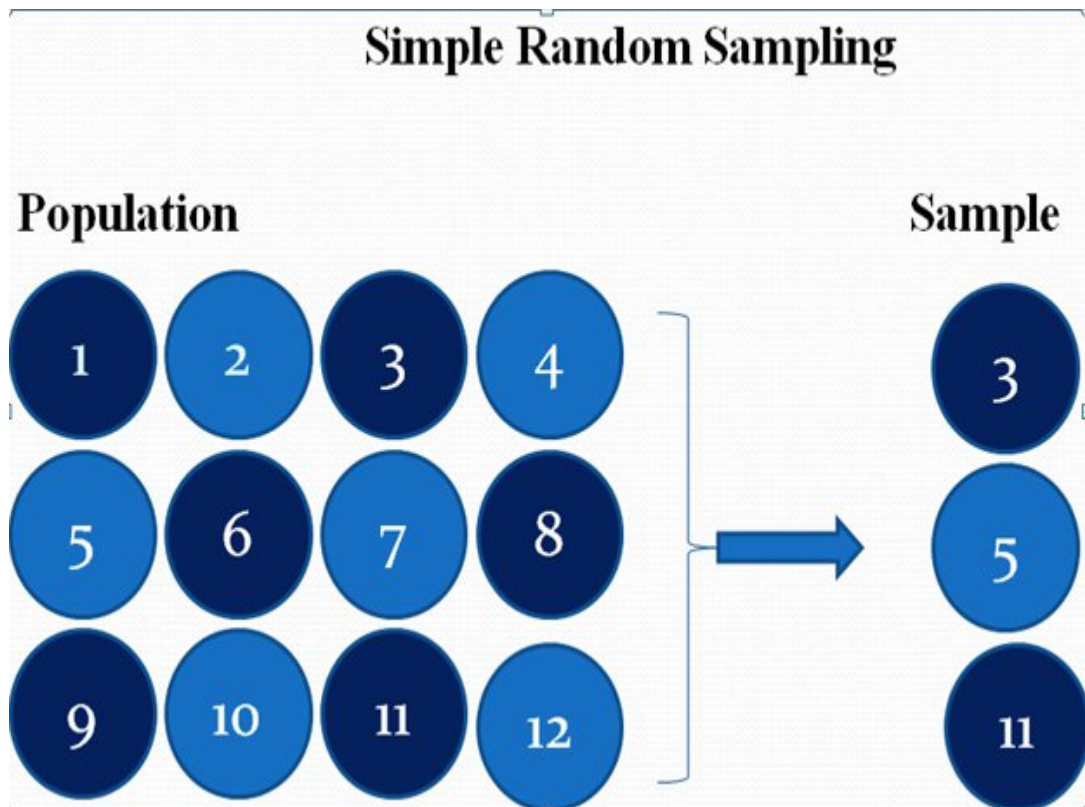
1. Probability Sampling Techniques

Probability sampling is defined by the principle that every member of the population has a known and equal chance of being selected. This method is critical for producing unbiased, representative samples.

a) Simple Random Sampling

Simple random sampling is the most straightforward probabilistic sampling technique. Every member of the population has an equal chance of being included in the sample, similar to a lottery. This method requires a complete list of the population, from which members are chosen randomly, either manually or using random number generation tools. Suppose you have a list of 10,000 voters in a town and you want to survey 1,000 of them. Each voter's name is assigned a number, and a random number generator picks 1,000 unique numbers corresponding to the voters who will be surveyed.





Simple Random Sampling- When to Use It

Simple random sampling is an appropriate choice under certain conditions or circumstances.

Homogeneous Population: Simple random sampling is most suitable when the population is relatively homogeneous, meaning that all individuals have similar characteristics or attributes. In such cases, each individual has an equal chance of being selected, ensuring a representative sample.

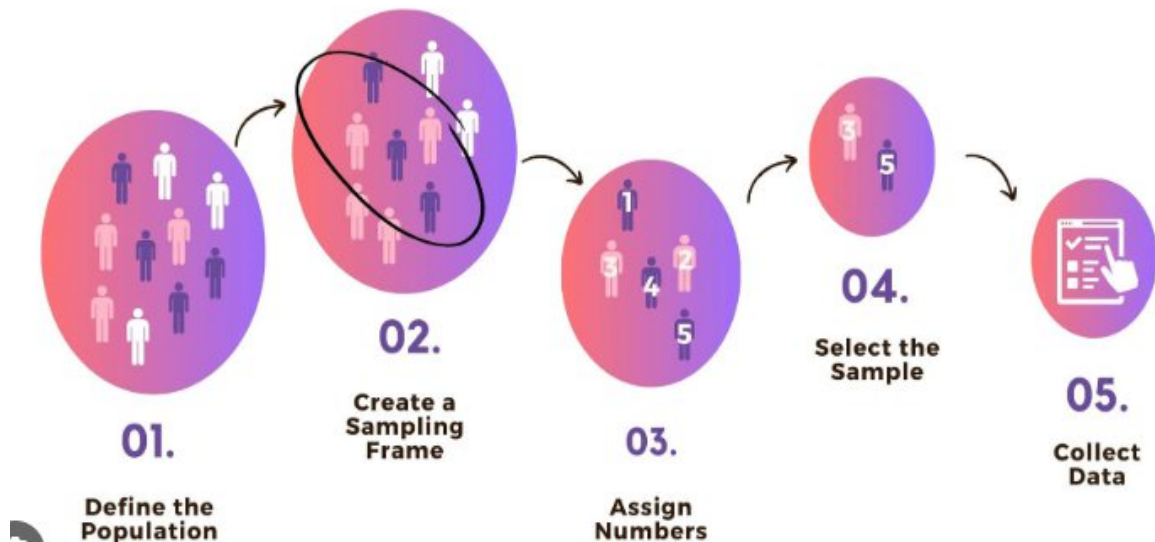
Small Population Size: When the population size is small and easily accessible, simple random sampling is feasible and efficient. It allows for the selection of a representative sample without the need for complex sampling methods.

Limited Resources: Simple random sampling is cost-effective and straightforward to implement compared to more complex sampling techniques. It requires minimal resources in terms of time, effort, and budget, making it ideal for studies with limited resources.

Equal Representation: If there are no specific subgroups or strata within the population that need to be oversampled or under-sampled, simple random sampling ensures that each subgroup is represented proportionally in the sample.

Population Heterogeneity: When the population exhibits homogeneity or random distribution of characteristics of interest, simple random sampling provides an unbiased representation of the population's diversity.

A Process of Simple Random Sampling



b) Stratified Sampling

Stratified sampling involves dividing the population into distinct subgroups or strata based on certain characteristics, such as age, gender, or income level. Samples are then randomly selected from each stratum in proportion to their representation in the population. This method ensures that each subgroup is adequately represented in the sample, making it useful when certain subgroups are of particular interest or importance.

For example, you're conducting a survey on smartphone usage habits among teenagers in a city. You know that the population of teenagers is diverse in terms of school grades (e.g., 9th, 10th, 11th, and 12th grades). You decide to divide the population into four strata based on school grade and then randomly select 100 teenagers from each grade stratum to ensure proportional representation in your sample.

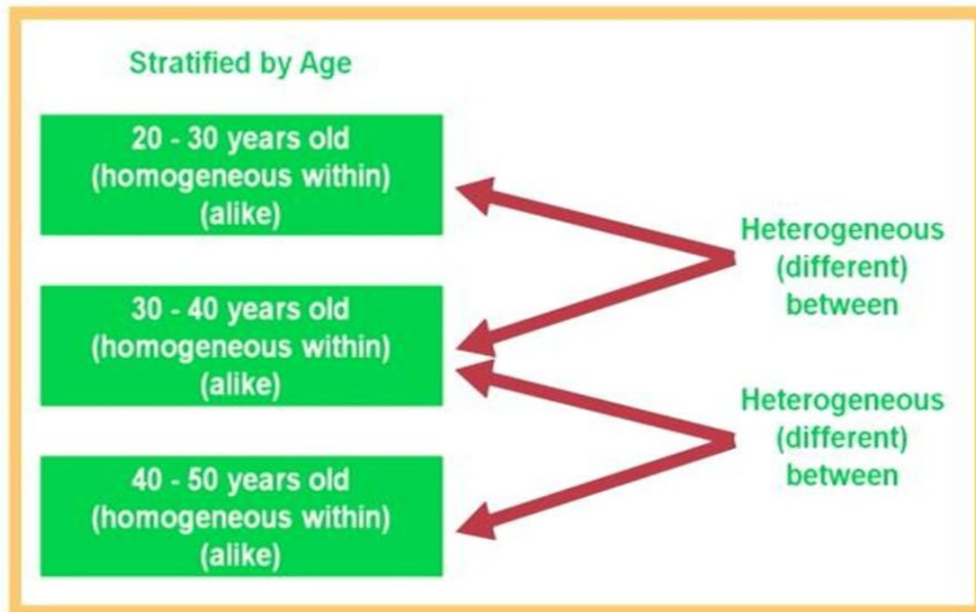
Stratified Sampling: When to Use It

Heterogeneous Population: When the population exhibits variability in characteristics of interest.

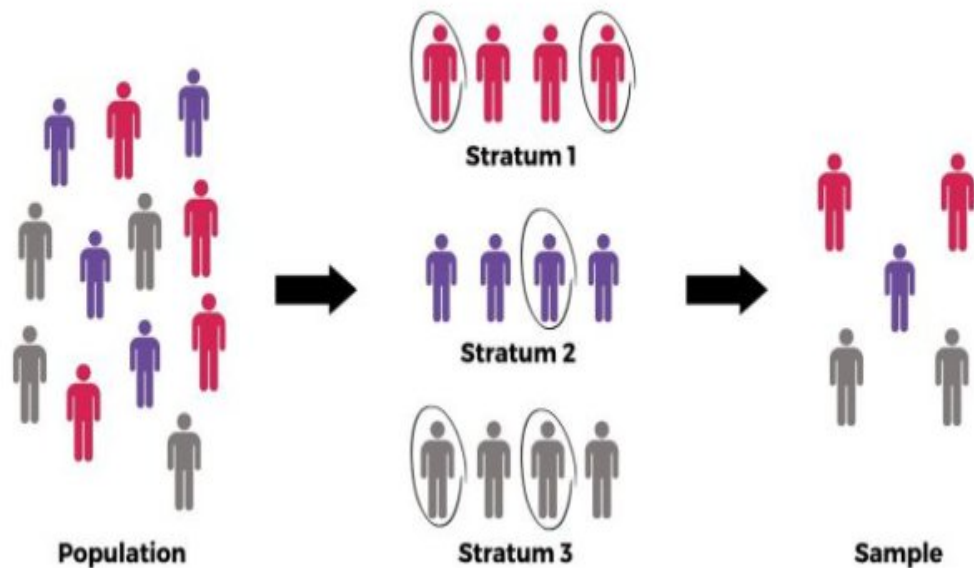
Specific Subgroup Analysis: When certain subgroups need to be represented accurately in the sample.

Increased Precision: When precision is desired for estimates within each subgroup.

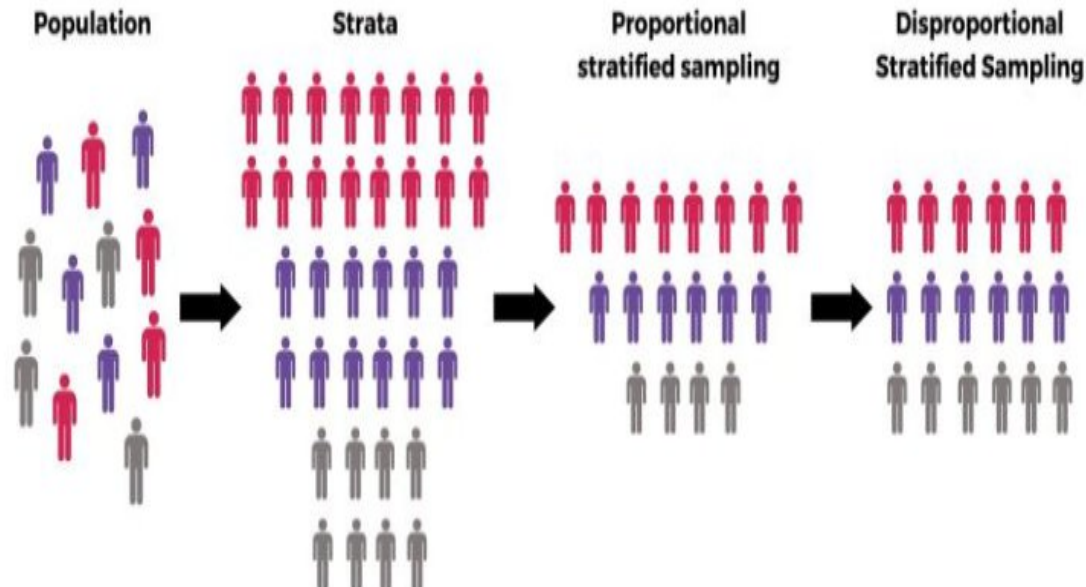
Stratified Random Sample: Population of FM Radio Listeners



Stratified Sampling



2 Types of Stratified Sampling



c) Cluster Sampling

Cluster sampling involves dividing the population into clusters or groups, such as households or schools, and then randomly selecting some clusters to include in the sample. All individuals within the selected clusters are then included in the sample. This method is efficient when it's difficult or impractical to obtain a complete list of individuals in the population, as clusters can be sampled more easily.

Consider a study aiming to assess the prevalence of obesity in rural communities. Instead of trying to survey every individual in each rural area, which might be impractical, you randomly select several villages or towns (clusters) and survey all individuals within those selected clusters. This method allows you to obtain a representative sample while reducing costs and logistical challenges.

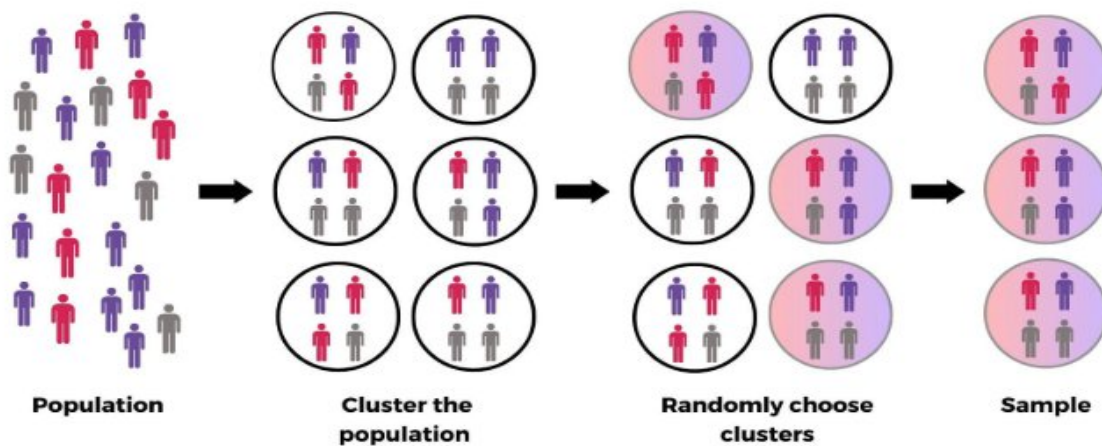
Cluster Sampling: When to Use It

Large Geographical Area: When the population is spread out over a large geographical area.

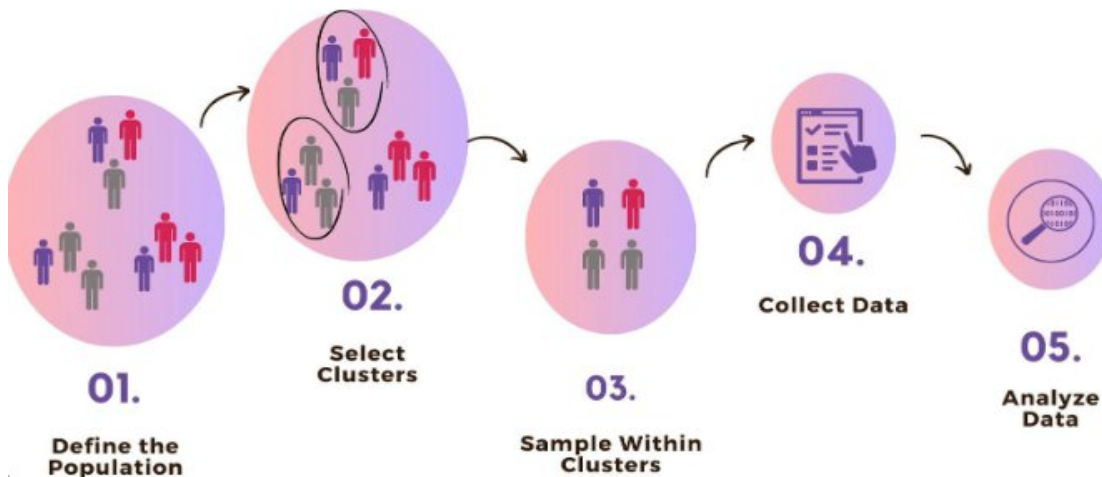
Cost and Time Efficiency: When it's more cost-effective and time-efficient to sample clusters rather than individuals.

Infeasibility of Complete Population List: When obtaining a complete list of individuals in the population is impractical.

Cluster Sampling



Steps of Cluster Sampling



d) Systematic Sampling

Systematic sampling involves selecting members from a larger population at a regular interval, determined by dividing the population size by the desired sample size. After randomly selecting a starting point within the first interval, the researcher selects every n th individual. For example, In an employee directory of 500 people, to conduct a systematic sample of 50 employees, you would sample every 10th person after selecting a random start between 1 and 10.

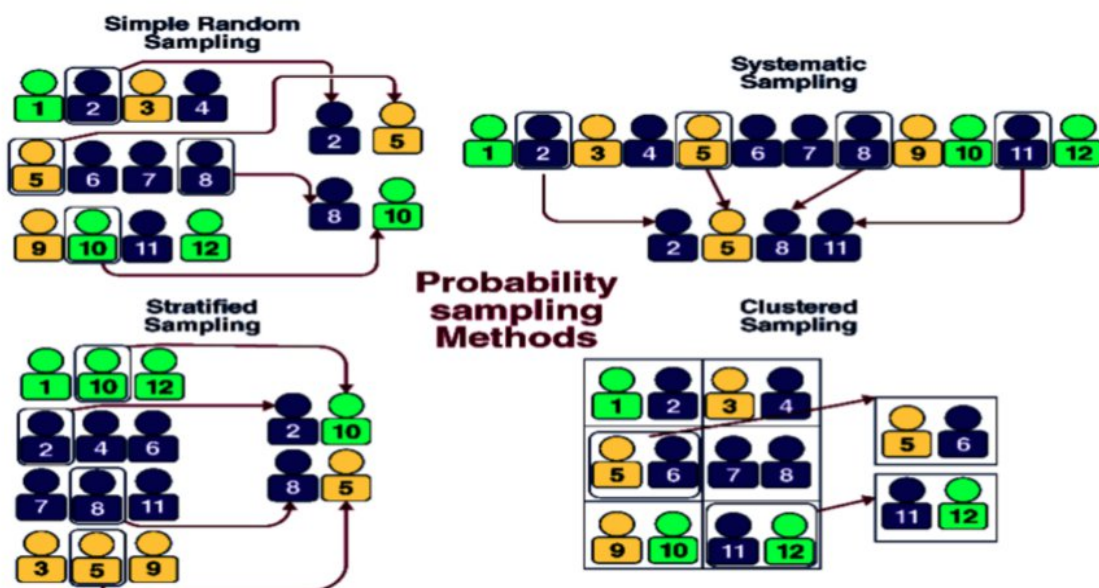
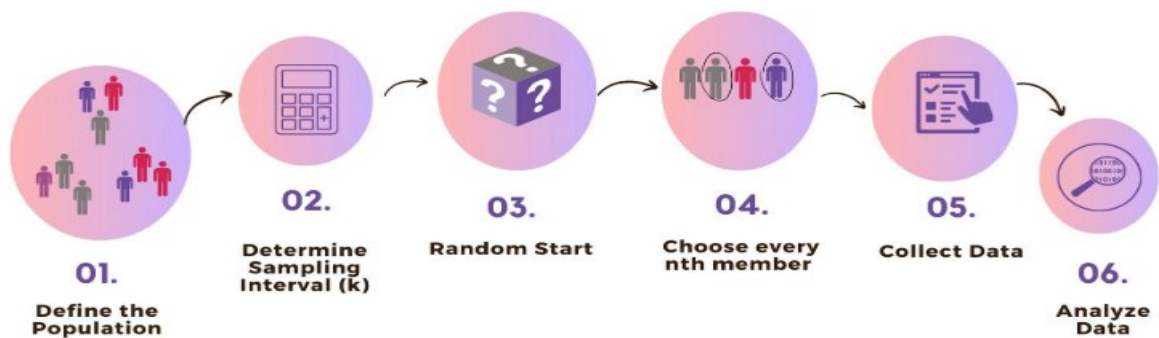
Systematic Sampling: When to Use It

- Large Populations
- Ordered Population
- Resource Constraints
- Accessibility and Convenience
- When Randomization is Difficult

Systematic Random Sampling



Steps to do Systematic Sampling

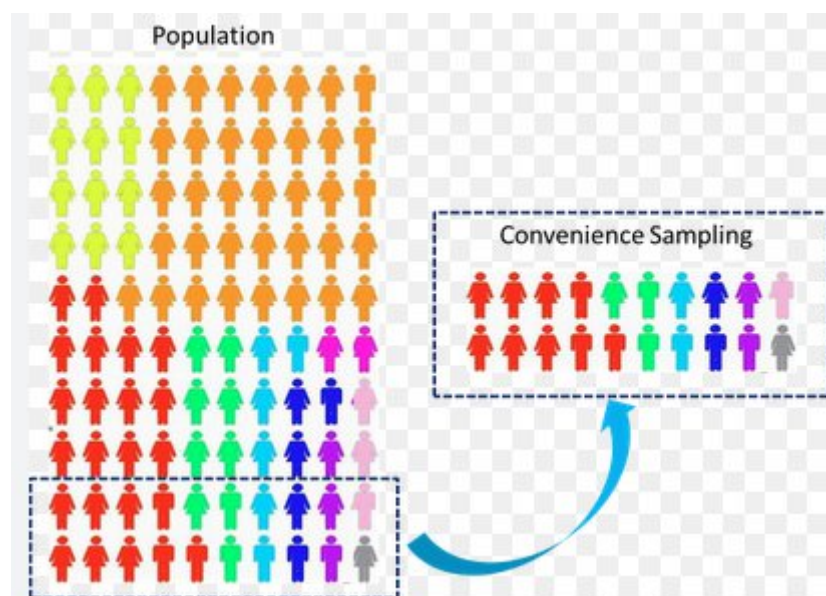


2. Non-Probability Sampling Techniques

Non-probability sampling methods do not provide all the members of the population an equal chance of participating in the study. These techniques are used when the availability of a complete list is not possible, or when the research does not require a random sample.

a) Convenience Sampling

Convenience sampling involves selecting individuals who are readily available or easily accessible to the researcher, rather than randomly selecting from the entire population. It is used when time, cost, or logistical constraints make it impractical to use other sampling methods. Convenience sampling is often employed in exploratory research, pilot studies, or when quick insights are needed, but it may introduce bias and limit the generalizability of findings.

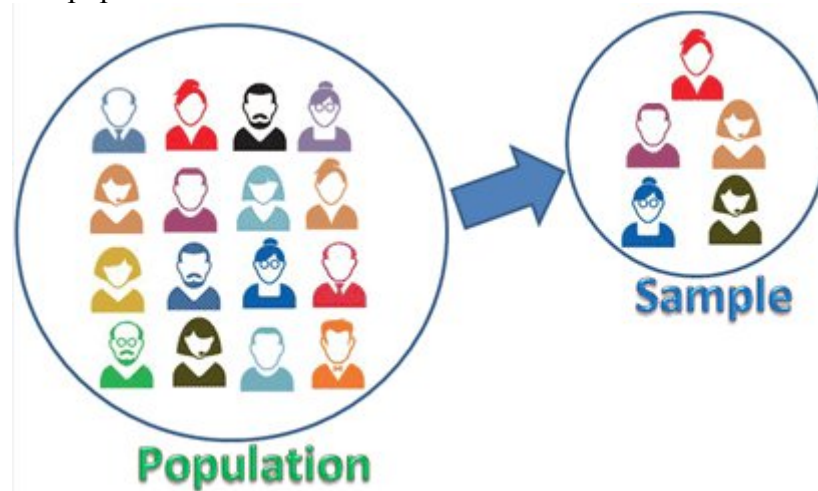


Steps to Use Convenience Sampling



b) Purposive/ Judgement Sampling

Purposive sampling involves selecting specific individuals or elements from a population based on predetermined criteria relevant to the research objectives. It is used when researchers seek to target particular characteristics or traits within the population, aiming to gain insights into specific subgroups or phenomena of interest. Purposive sampling is employed in situations where the researcher's expertise or prior knowledge guides the selection process, facilitating in-depth exploration or focused investigation of particular aspects within the population.

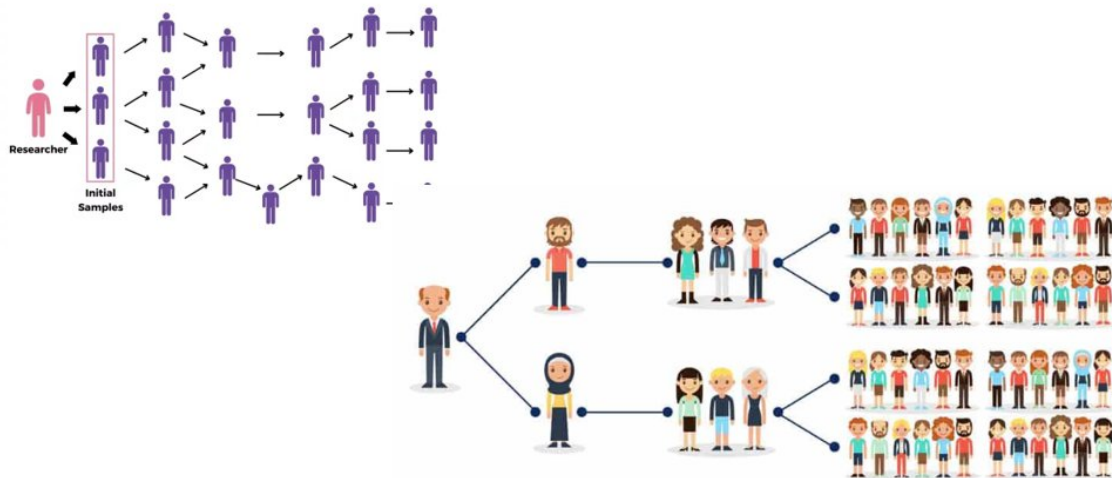


c) Snowball Sampling

Snowball sampling is a non-probability sampling method where existing participants recruit additional participants, typically through referrals. It is used when studying hard-to-reach or hidden populations, such as marginalized communities or individuals with rare characteristics.

Snowball sampling is advantageous in situations where traditional sampling methods may be impractical due to the lack of a sampling frame or difficulty in accessing the Population of interest.

Snowball sampling

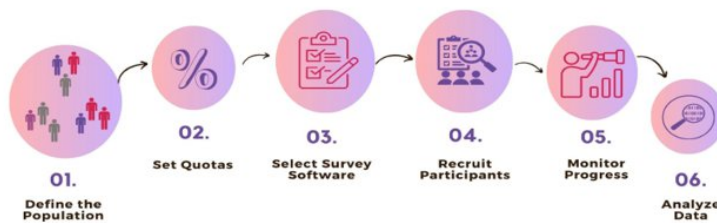


d) Quota Sampling

Quota sampling involves selecting a sample based on predetermined quotas to ensure representation of specific characteristics, such as age, gender, or socioeconomic status, in the sample. It is used when researchers need to ensure proportional representation of key population segments, but random sampling is impractical or not feasible. Quota sampling is often employed in market research, opinion polling, and social sciences research to obtain a sample that mirrors the demographic composition of the population being studied.



6 Steps To Do Proportional Quota Sampling

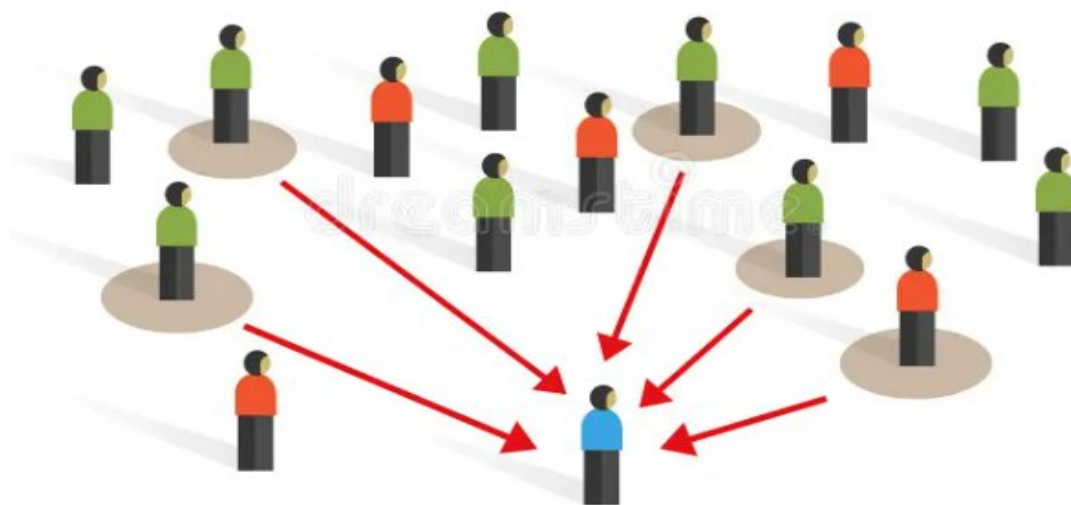


5. Voluntary response sampling

voluntary response sample consists of members of a population who volunteer to participate in a study. Like a convenience sample, a voluntary response sample is often based on convenient access to a population. However, instead of the researcher selecting participants, participants volunteer on their own. For example, let's say college administrators want to know how students feel about the food served on campus. They email students a link to an online survey about the quality of the food, and ask students to fill out the survey if they have time.

Voluntary response samples tend to suffer from nonresponse bias, which occurs when certain groups of people are less likely to provide responses. People who voluntarily respond will likely have stronger opinions, either positive or negative, than the rest of the population. In this case, only students who really like or really dislike the food may be motivated to fill out the survey. The survey may omit many students who have more mild opinions about the food, or are neutral. This makes the volunteer students unrepresentative of the overall student population.

Voluntary response sampling



UNIT-III

Topics: Hypothesis testing: Importance of Hypothesis testing, null and alternative hypotheses, Type- I and Type –II errors, approaches to hypothesis testing, Central limit theorem and its application, confidence interval. **Analysis of Variance**

(ANOVA): Introduction to ANOVA, one way ANOVA, two-way ANOVA.

Introduction

- **Hypothesis testing** is the often used strategy for deciding whether a sample data offer such support for a hypothesis that generalization can be made.
- Thus hypothesis testing enables us to make probability statements about population parameter (s).
- The hypothesis may not be proved absolutely, but in practice it is accepted if it has withstood a critical testing.

Hypothesis Examples

A hypothesis predicts the relationship between the independent and dependent variable.

There are different ways of stating a hypothesis.

NULL HYPOTHESIS	SIMPLE HYPOTHESIS	NONDIRECTIONAL HYPOTHESIS
The number of calories consumed has no effect on weight.	If you consume fewer calories, you'll lose weight.	There is a relationship between calories consumed and weight.

Background image: A scale of calories with a needle pointing to 100. The word 'CALORIES' is written on the scale.

sciencenotes.org

Hypothesis: when one talks about hypothesis, one simply means a mere assumption or some supposition to be proved or disproved. But for a researcher hypothesis is a formal question that he intends to resolve.

- Thus a hypothesis may be defined as a proposition or a set of proposition set forth as an explanation for the occurrence of some specified group of phenomena,
- either asserted merely as a provisional conjecture to guide some investigation or accepted as highly probable in the light of established facts.
- Quite often a research hypothesis is a predictive statement, capable of being tested by scientific methods, that relates an independent variable to some dependent variable.

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- Ex: consider statements like the following ones: “Students who receive counseling will show a greater
 - increase in creativity than students not receiving counseling”
- Or
- “the automobile A is performing as well as automobile B.” These are hypotheses capable of being objectively verified and tested.
 - Thus, we may conclude that a hypothesis states what we are looking for and it is a proposition which can be put to a test to determine its validity.

Importance of Hypothesis

Hypothesis should be clear and precise. If the hypothesis is not clear and precise, the inferences drawn on its basis cannot be taken as reliable.

- It ensures the entire research methodologies are scientific and valid.
- It helps to assume the probability of research failure and progress.

- It helps to provide link to the **underlying theory and specific research question.**
- It helps in data analysis and measure **the validity and reliability of the research.**
- It provides a basis or **evidence to prove the validity of the research.**
- It helps to describe **research study in concrete terms rather than theoretical terms.**

Characteristics of Good Hypothesis:

- Should be simple.
- Should be specific.
- Should be stated in advance.

FLOW DIAGRAM FOR HYPOTHESIS TESTING

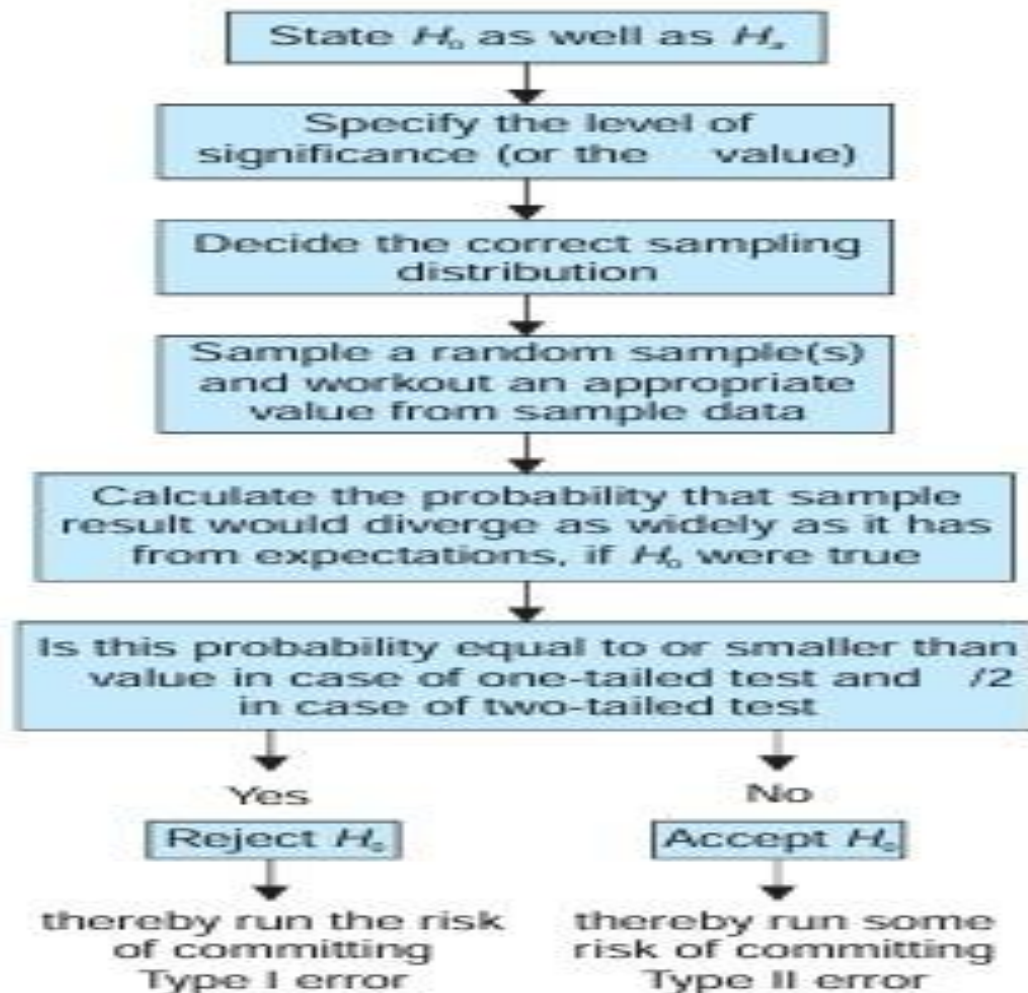


Fig. 9.4

- To test a hypothesis means to tell (on the basis of the data the researcher has collected) **whether or not the hypothesis seems to be valid.**
- In hypothesis testing the main question is: **whether to accept the null hypothesis or not to accept the null hypothesis?**
- Procedure for hypothesis testing refers to all those steps that we undertake for making a choice between the two actions i.e., rejection and acceptance of a null hypothesis.
- Basic concepts in the context of **testing of hypotheses** need to be explained.
- **(a) Null hypothesis and alternative hypothesis:** In the context of statistical analysis, we often talk about null hypothesis and alternative hypothesis.
- **Null Hypothesis:** If we are to compare method A with method B about its superiority and if we proceed on the assumption that both methods are equally good, **then this assumption is termed as the null hypothesis.**
- **Alternative Hypothesis:** As against this, we may think that the method A is superior or the method B is inferior, we are then stating **what is termed as alternative hypothesis.**
- The null hypothesis is generally symbolized as H_0 and the alternative hypothesis as H_a . Suppose we want to test the hypothesis that the population mean μ is equal to the hypothesized mean $\mu = \mu_{H_0} = 100$.
- Then we would say that the null hypothesis is that the population mean is equal to the hypothesized mean 100 and symbolically we can express as:

$$H_0 : \mu = \mu_{H_0} = 100$$

Alternative Hypothesis: we should conclude that something else is true. What we conclude rejecting the null hypothesis is known as alternative hypothesis.

- In other words, the set of alternatives to the null hypothesis is referred to as the alternative hypothesis. If we accept H_0 , then we are rejecting H_a and if we reject H_0 , then we are accepting H_a . For
- we may consider three possible alternative hypotheses as follows*:
- The null hypothesis and the alternative hypothesis are chosen before the sample is drawn (the researcher must avoid the error of deriving hypotheses from the data that he collects and then testing

$$H_0 : \mu = \mu_{H_0} = 100$$

<i>Alternative hypothesis</i>	<i>To be read as follows</i>
$H_a : \mu \neq \mu_{H_0}$	(The alternative hypothesis is that the population mean is not equal to 100 i.e., it may be more or less than 100)
$H_a : \mu > \mu_{H_0}$	(The alternative hypothesis is that the population mean is greater than 100)
$H_a : \mu < \mu_{H_0}$	(The alternative hypothesis is that the population mean is less than 100)

the hypotheses from the same data).

In the choice of null hypothesis, the following considerations are usually kept in view:

- (a) **Alternative hypothesis is usually the one which one wishes to prove and the null hypothesis is the one which one wishes to disprove.** Thus, a null hypothesis represents the hypothesis we are trying to reject, and alternative hypothesis represents all other possibilities.

(b) If the rejection of a certain hypothesis when it is actually true involves great risk, it is taken as null hypothesis because then the probability of rejecting it when it is true is α (the level of significance) which is chosen very small.

(c) Null hypothesis should always be specific hypothesis i.e., it should not state about or approximately a certain value.

Type 1 and Type2 Errors

This concept of testing basically two types of errors we can make.

1. Type 1 error

2. Type 2 error

Type 1 Error: We may reject H_0 when H_0 is true. [Rejection hypothesis which should have been accepted]

Type 2 error: We may Accept H_0 when in fact H_0 is not true.[Accepting the hypothesis which should have been rejected]

Table 9.2

	Decision	
	Accept H_0	Reject H_0
H_0 (true)	Correct decision	Type I error (α error)
H_0 (false)	Type II error (β error)	Correct decision

Outcomes and Probabilities

Possible Hypothesis Test Outcomes		
Decision	State of Nature	
	H_0 True	H_0 False
	Do Not Reject H_0 No error ($1 - \alpha$)	Type II Error (β)
Reject H_0	Type I Error (α)	No Error ($1 - \beta$)

Key:
Outcome
(Probability)

b) The level of significance: This is a very important concept in the context of hypothesis testing. It is always some percentage (usually 5%) which should be chosen with great care, thought and reason. In case we take the significance level at 5 per cent, then this implies that H_0 will be rejected.

- when the sampling result (i.e., observed evidence) has a less than 0.05 probability of occurring if H_0 is true. In other words, the 5 per cent level of significance means that researcher is willing to take as much as a 5 per cent risk of rejecting the null hypothesis when it (H_0) happens to be true.
- Thus the significance level is the maximum value of the probability of rejecting H_0 when it is true and is usually determined in advance before testing the hypothesis.
- (c) **Decision rule or test of hypothesis:** *Given a hypothesis H_0 and an alternative hypothesis H_a , we make a rule which is known as decision rule according to which we accept H_0 (i.e., reject H_a) or reject H_0 (i.e., accept H_a).*
- For instance, if (**H_0 is that a certain lot is good (there are very few defective items in it) against H_a**) **that the lot is not good (there are too many defective items in it)**, then we must decide the number of items to be tested and the criterion for accepting or rejecting the hypothesis.
- *We might test 10 items in the lot and plan our decision saying that if there are none or only 1 defective item among the 10, we will accept H_0 otherwise we will reject H_0 (or accept H_a). This sort of basis is known as decision rule.

d) Type I and Type II errors: In the context of testing of hypotheses, there are basically two types of errors we can make.

- We may reject H_0 when H_0 is true and we may accept H_0 when in fact H_0 is not true. The former is known as Type I error and the latter as Type II error.

In other words, Type I error means rejection of hypothesis which should have been accepted and Type II error means accepting the hypothesis which should have been rejected.

Type I error is denoted by α (alpha) known as α error, also called the level of significance of test; and

Type II error is denoted by β (beta) known as β error

NONPARAMETRIC TESTING

Non-parametric test is a statistical analysis method that **does not assume the population data belongs to some prescribed distribution**, which is determined by some parameters. Due to this, a **non-parametric test is also known as a distribution-free test**. These tests are usually based on distributions that have unspecified parameters.

PARAMETRIC TESTING

Parametric tests are statistical measures used in the analysis phase of research to draw inferences and conclusions to solve a research problem. There are various types of parametric tests, **such as z-test, t-test and F-test**. The selection of a particular test for research depends upon various factors, such as the type of population, sample size, Standard Deviation (SD) and variance of population. **It is important for a researcher to identify the appropriate test to maintain the authenticity and validity of research results.**

The important parametric tests are: (1) z-test; (2) t-test; (3) χ^2 -test, and (4) F-test.

All these tests are based on the assumption of normality i.e., the source of data is considered to be normally distributed.

z-test

- It is based on the normal probability distribution and is used for judging the significance of several statistical measures, particularly the mean.
- The relevant test statistic*, z , is worked out and compared with its probable value (to be read from table showing area under normal curve) at a specified level of significance for judging the significance of the measure concerned.
- z -test is generally used for comparing the mean of a sample to some hypothesized mean for the population in case of large sample, or when population variance is known.
- **z -test is also used for judging the significance of difference between means of two independent samples in case of large samples, or when population variance is known.**

t - Test

- t -test is based on t -distribution and is considered an appropriate test for judging the significance of a sample mean or for judging the significance of difference between the means of two samples in case of small sample(s).
- when population variance is not known (in which case we use variance of the sample as an estimate of the population variance).
- In case two samples are related, we use paired t -test (or what is known as difference test) for judging the significance of the mean of difference between the two related samples.
- It can also be used for judging the significance of the coefficients of simple and partial correlations.

χ^2 -test

- χ^2 -test is based on chi-square distribution and as a parametric test is used for comparing a sample variance to a theoretical population variance.

F-test

F-test is based on F-distribution and is used to compare the variance of the two-independent samples. This test is also used in the context of analysis of variance (ANOVA) for judging the significance of more than two sample means at one and the same time.

It is also used for judging the significance of multiple correlation coefficients.

Test statistic, F, is calculated and compared with its probable value for accepting or rejecting the null hypothesis.

- Mean of the population can be tested presuming different situations such as the population may be normal or other than normal, it may be finite or infinite, sample size may be large or small, variance of the population may be known or unknown and the alternative hypothesis may be two-sided or one sided. Our testing technique will differ in different situations. We may consider some of the important situations. 1. Population normal, population infinite, sample size may be large or small but variance of the population is known, H_a may be one-sided or two-sided: In such a situation z-test is used for testing hypothesis of mean and the test statistic z is worked out as under:

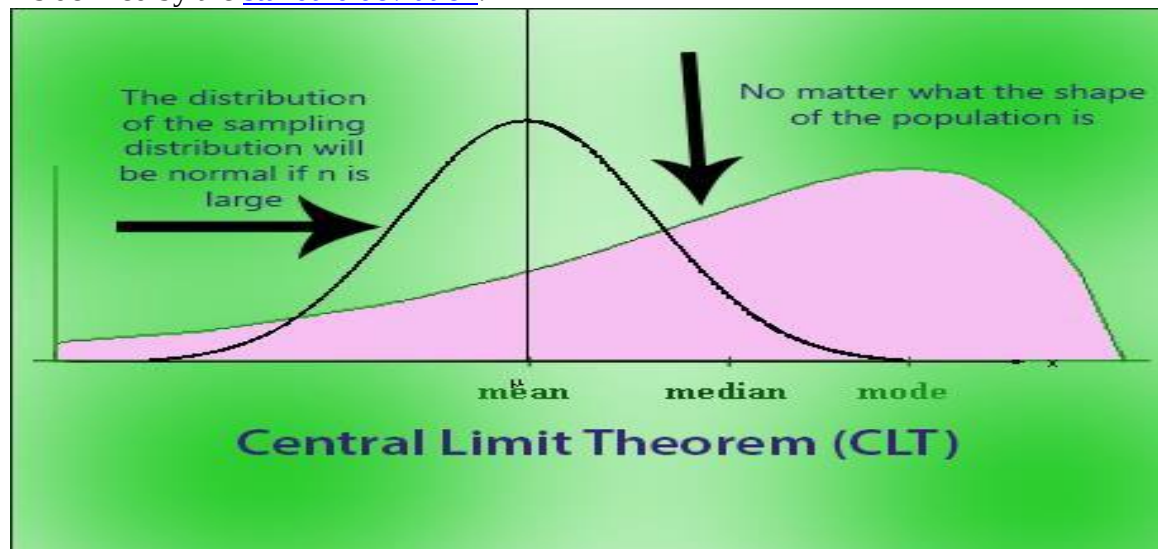
CLT

- The Central Limit Theorem (CLT) is a fundamental concept in statistics that states that the distribution of the sample means of a population will be approximately **normally distributed**, regardless of the original distribution of the population, given that the sample size is sufficiently large.
- “Normally distributed” means that data is symmetrically distributed around a central value, with most values falling close to the center. This distribution is also known as a **Gaussian Distribution Function or bell curve**. The normal distribution is the most common type of distribution assumed in technical stock market analysis.

The standard [normal distribution has two parameters](#): the mean and the standard deviation.

In distribution, [mean](#) (average), [median](#) (midpoint), and [mode](#) (most frequent observation) are equal.

These values represent the peak or highest point. The distribution then falls symmetrically around the mean, the width of which is defined by the [standard deviation](#).



Central Limit Theorem Formula

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Sample Mean = Population Mean = μ

Sample Standard Deviation = $\frac{\text{Standard Deviation}}{\sqrt{n}}$

OR

Sample Standard Deviation = $\frac{\sigma}{\sqrt{n}}$

RTA with CLT on Public Opinion Surveys

- **Political Polling:** Polling groups conduct surveys to see what people think about politicians, laws, and issues. They make sure their surveys are accurate by asking a diverse group of people. They use random sampling techniques to do this. Using the Central Limit Theorem (CLT), they can figure out the margin of error for their estimates and how people in the whole population might respond. This helps them predict election results and understand what people think about different policies. It gives useful information to politicians, decision-makers.
- **Market Research:** Businesses often use surveys to learn what customers think about their products, services, and brands. They might ask questions to understand what features people want in a product or how satisfied customers are with their brand. Using the Central Limit Theorem (CLT), researchers can estimate what most people in the target market think by gathering data from a representative group of customers. This helps businesses make better decisions about how to manage customer relationships, market their products, and develop new ones, making them more competitive.

RTA with CLT on Meteorology:

- **Weather Forecasting:** Meteorologists use real climate data to predict future weather patterns and events. They collect data from various weather stations over specific time periods to calculate average values and variations for factors like temperature, humidity, wind speed, and air pressure. Using the Central Limit Theorem (CLT), meteorologists can estimate how these weather factors are distributed across a region and make probabilistic forecasts for future conditions. For example, they might predict the likelihood of rain, the range of temperatures expected, or the intensity of wind speeds for a certain period. These forecasts help individuals, groups, and government agencies prepare for potential weather-related risks and adjust their plans accordingly.
- **Climate Modeling:** Climate scientists use computer models to simulate Earth's climate and study long-term patterns and changes. These models use data from observations and educated guesses for various weather factors like temperature, rainfall, sea level, and ice coverage. By adjusting the settings and input data, scientists can create collections of climate predictions showing possible future conditions. They use the Central Limit Theorem (CLT) to analyze the results of these simulations by calculating average values and variations for important climate indicators. This helps them understand the uncertainty surrounding climate projections and provide clear, informed views on potential climate impacts and adaptation measures.

Problems on Unit 3:

7Q. Set up an analysis of variance table for the following per acre production data for three varieties of wheat, each grown on 4 plots and state if the variety differences are significant.

Per acre production data			
Plot of land	Variety of wheat		
	A	B	C
1	6	5	5
2	7	5	4
3	3	3	3
4	8	7	4

[critical value F (at 5% LOS) $F(2,9) = 4.26$] Ans

Solution: We wish to test
 H_0 : All wheat varieties are the same or effects of different varieties on wheat production are the same
 H_1 : All varieties are not the same

First we calculate the mean of each of these samples:

$$\bar{x}_1 = \frac{6+7+3+8}{4} = 6$$

$$\bar{x}_2 = \frac{5+5+3+7}{4} = 5$$

$$\bar{x}_3 = \frac{5+4+3+4}{4} = 4$$

Mean of the sample means or

$$\begin{aligned}\bar{\bar{x}} &= \frac{\bar{x}_1 + \bar{x}_2 + \bar{x}_3}{k} \\ &= \frac{6 + 5 + 4}{3} = 5\end{aligned}$$

Now we work out SS between and SS within samples:

$$\begin{aligned}SS \text{ between} &= n_1 (\bar{x}_1 - \bar{\bar{x}})^2 + n_2 (\bar{x}_2 - \bar{\bar{x}})^2 + n_3 (\bar{x}_3 - \bar{\bar{x}})^2 \\ &= 4(6 - 5)^2 + 4(5 - 5)^2 + 4(4 - 5)^2 \\ &= 4 + 0 + 4 \\ &= 8\end{aligned}$$

$$\begin{aligned}SS \text{ within} &= \sum (x_{1i} - \bar{x}_1)^2 + \sum (x_{2i} - \bar{x}_2)^2 + \sum (x_{3i} - \bar{x}_3)^2; \quad i = 1, 2, 3, 4 \\ &= \{(6 - 6)^2 + (7 - 6)^2 + (3 - 6)^2 + (8 - 6)^2\} \\ &\quad + \{(5 - 5)^2 + (5 - 5)^2 + (3 - 5)^2 + (7 - 5)^2\} \\ &\quad + \{(5 - 4)^2 + (4 - 4)^2 + (3 - 4)^2 + (4 - 4)^2\} \\ &= \{0 + 1 + 9 + 4\} + \{0 + 0 + 4 + 4\} + \{1 + 0 + 1 + 0\} \\ &= 14 + 8 + 2 \\ &= 24\end{aligned}$$

$$\begin{aligned}SS \text{ for total variance} &= \sum (x_{ij} - \bar{\bar{x}})^2 \quad i = 1, 2, 3 \dots \text{ and } j = 1, 2, 3 \dots \\ &= (6 - 5)^2 + (7 - 5)^2 + (3 - 5)^2 + (8 - 5)^2 \\ &\quad + (5 - 5)^2 + (5 - 5)^2 + (3 - 5)^2 \\ &\quad + (7 - 5)^2 + (5 - 5)^2 + (4 - 5)^2 \\ &\quad + (3 - 5)^2 + (4 - 5)^2\end{aligned}$$

$$= 1 + 4 + 4 + 9 + 0 + 0 + 4 + 4 + 0 + 1 + 4 + 1$$

$$= 32$$

Alternatively, it (SS for total variance) can also be worked out thus:

SS for total = SS between + SS within

$$= 8 + 24 = 32$$

We can now set up the ANOVA table for this problem:

Table 12.3

Source of variation	SS	d.f.	MS	F-ratio	Critical value F(at 5%) (from the F-table)
Between sample	8	$(3 - 1) = 2$	$8/2 = 4.00$	$4.00/2.67 = 1.5$	$F(2, 9) = 4.26$
Within sample	24	$(12 - 3) = 9$	$24/9 = 2.67$		
Total	32	$(12 - 1) = 11$			

The above table shows that the calculated value of F is 1.5 which is less than the table value of 4.26 at 5% level with d.f. being $v_1 = 2$ and $v_2 = 9$ and hence could have arisen due to chance. This analysis supports the null-hypothesis of no difference in means. We may, therefore, conclude that the difference in wheat output due to varieties is insignificant and is just a matter of chance.

8Q. Set up an analysis of variance table for the following two-way design results

Per acre production data of Wheat

Varieties of Fertilizers	Variety of Seeds		
	A	B	C
W	6	5	5
X	7	5	4
Y	3	3	3
Z	8	7	4

Also state whether variety differences are significant at 5 % level [Critical value $F(2,6) = 5.14$, $F(3,6) = 4.76$]

Ans

Step (i) $T = 60, n = 12, \therefore \text{Correction factor} = \frac{(T)^2}{n} = \frac{60 \times 60}{12} = 300$

Step (ii) $\text{Total SS} = (36 + 25 + 25 + 49 + 25 + 16 + 9 + 9 + 9 + 64 + 49 + 16) - \left(\frac{60 \times 60}{12} \right)$
 $= 332 - 300$
 $= 32$

Step (iii) $\text{SS between columns treatment} = \left[\frac{24 \times 24}{4} + \frac{20 \times 20}{4} + \frac{16 \times 16}{4} \right] - \left[\frac{60 \times 60}{12} \right]$
 $= 144 + 100 + 64 - 300$
 $= 8$

Step (iv) $\text{SS between rows treatment} = \left[\frac{16 \times 16}{3} + \frac{16 \times 16}{3} + \frac{9 \times 9}{3} + \frac{19 \times 19}{3} \right] - \left[\frac{60 \times 60}{12} \right]$
 $= 85.33 + 85.33 + 27.00 + 120.33 - 300$
 $= 18$

Step (v) $\text{SS residual or error} = \text{Total SS} - (\text{SS between columns} + \text{SS between rows})$
 $= 32 - (8 + 18) = 6$

TABLE 12.8: THE ANOVA TABLE

Source of variation	SS	d.f.	MS	F-ratio	5% F-limit (or the tables values)
Between columns (i.e., between varieties of seeds)	8	$(3 - 1) = 2$	$8/2 = 4$	$4/1 = 4$	$F(2, 6) = 5.14$
Between rows (i.e., between varieties of fertilizers)	18	$(4 - 1) = 3$	$18/3 = 6$	$6/1 = 6$	$F(3, 6) = 4.76$
Residual or error	6	$(3 - 1) \times (4 - 1) = 6$	$6/6 = 1$		
Total	32	$(3 \times 4) - 1 = 11$			

ANOVA table can be set up for the given problem as shown in Table 12.8.

From the said ANOVA table, we find that differences concerning varieties of seeds are insignificant at 5% level as the calculated F -ratio of 4 is less than the table value of 5.14, but the variety differences concerning fertilizers are significant as the calculated F -ratio of 6 is more than its table value of 4.76.

9Q. Explain the process of Two-Way ANOVA in detail. Ans

Two way Anova

- Two-way ANOVA involves two independent variables
- Compares the means of groups based on two or more factors
- Used to assess the impact of both variables and their interaction
- For example, analyzing the effect of diet and exercise on weight loss

It is used when the data are classified on the basis of two factors.

Steps to solve two way Anova problems:

Step 1: Take the total of observations and call it T

Step 2: Correction factor= $(T^2)/n$

Step 3: Calculate SST(Total sum of square)

$$\text{Total SS} = \sum x_{ij}^2 - \frac{(T)^2}{n}$$

Step 4: Calculate SS between the columns:

- 1) Take total of different columns
- 2) Obtain square of each column total
- 3) Divide these squared values of each column by no of items in each column
- 4) Obtain the total and subtract correction factor

Step 5: Calculate SS between the rows:

- 1) Take total of different rows
- 2) Obtain square of each row total
- 3) Divide these squared values of each row by no of items in each row
- 4) Obtain the total and subtract correction factor

Step 6: Calculate sum of square of deviation for residual or error variance

SS for residual or error variance = Total SS - SS between columns - SS between rows

Step 7: degree of freedom can be calculated as

Df for total variance= $(c \cdot r - 1)$

Df for variance between columns = $(c-1)$

Df for variance between rows = $(r-1)$

Df for residual variance= $(c-1)(r-1)$ Where

c = Number of columns

R= Number of rows

Step 8: : Complete the following table:

Summary Table:

Source of variation	Sum of squares (SS)	Degrees of freedom (d.f.)	Mean square (MS)	F-ratio
Between columns treatment	$\sum \frac{(T_j)^2}{n_j} - \frac{(T)^2}{n}$	$(c - 1)$	$\frac{SS \text{ between columns}}{(c - 1)}$	$\frac{MS \text{ between columns}}{MS \text{ residual}}$
Between rows treatment	$\sum \frac{(T_i)^2}{n_i} - \frac{(T)^2}{n}$	$(r - 1)$	$\frac{SS \text{ between rows}}{(r - 1)}$	$\frac{MS \text{ between rows}}{MS \text{ residual}}$
Residual or error	Total SS - (SS between columns + SS between rows)	$(c - 1)(r - 1)$	$\frac{SS \text{ residual}}{(c - 1)(r - 1)}$	
Total	$\sum x_{ij}^2 - \frac{(T)^2}{n}$	$(c \cdot r - 1)$		

Step 9: Compare the calculated F Ratio with table value and make the decision

If calculated value > table value Reject the null Hypothesis

Else accept the Null hypothesis

10Q. Set up ANOVA table for the following information relating to three drugs testing to judge the effectiveness in reducing blood pressure for three different groups of people.

Amount of Blood Pressure Reduction in Millimetres of Mercury

Group of People	Drug		
	X	Y	Z
A	14	10	11
	15	9	11
B	12	7	10
	11	8	11
C	10	11	8
	11	11	7

Do the drug act differently?

Are the different group of people affected differently? Is the interaction term significant?

Answers the above questions taking significant level of 5 % [critical value

$F(2,9) = 4.26$, $F(4,9) = 3.63$]

Ans

Step (i) $T = 187, n = 18$, thus, the correction factor $= \frac{187 \times 187}{18} = 1942.72$

Step (ii) Total $SS = [(14)^2 + (15)^2 + (12)^2 + (11)^2 + (10)^2 + (11)^2 + (10)^2 + (9)^2 + (7)^2 + (8)^2 + (11)^2 + (11)^2 + (11)^2 + (11)^2 + (10)^2 + (11)^2 + (8)^2 + (7)^2] - \left[\frac{(187)^2}{18} \right]$

$$= (2019 - 1942.72)$$

$$= 76.28$$

Step (iii) SS between columns (i.e., between drugs) $= \left[\frac{73 \times 73}{6} + \frac{56 \times 56}{6} + \frac{58 \times 58}{6} \right] - \left[\frac{(187)^2}{18} \right]$

$$= 888.16 + 522.66 + 560.67 - 1942.72$$

$$= 28.77$$

Step (iv) SS between rows (i.e., between people) $= \left[\frac{70 \times 70}{6} + \frac{59 \times 59}{6} + \frac{58 \times 58}{6} \right] - \left[\frac{(187)^2}{18} \right]$

$$= 816.67 + 580.16 + 560.67 - 1942.72$$

$$= 14.78$$

Step (v) SS within samples $= (14 - 14.5)^2 + (15 - 14.5)^2 + (10 - 9.5)^2 + (9 - 9.5)^2 + (11 - 11)^2 + (11 - 11)^2 + (12 - 11.5)^2 + (11 - 11.5)^2 + (7 - 7.5)^2 + (8 - 7.5)^2 + (10 - 10.5)^2 + (11 - 10.5)^2 + (10 - 10.5)^2 + (11 - 10.5)^2 + (11 - 11)^2 + (11 - 11)^2 + (8 - 7.5)^2 + (7 - 7.5)^2$

$$= 3.50$$

Step (vi) SS for interaction variation $= 76.28 - [28.77 + 14.78 + 3.50]$

$$= 29.23$$

Source of variation	SS	d.f.	MS	F-ratio	Critical at 5%
Between columns (i.e., between drugs)	28.77	$(3 - 1) = 2$	$\frac{28.77}{2}$ $= 14.385$	$\frac{14.385}{0.389}$ $= 36.9$	$F(2, 9) = 4.26$
Between rows (i.e., between people)	14.78	$(3 - 1) = 2$	$\frac{14.78}{2}$ $= 7.390$	$\frac{7.390}{0.389}$ $= 19.0$	$F(2, 9) = 4.26$
Interaction	29.23	4	$\frac{29.23}{4}$ $= 7.308$	$\frac{7.308}{0.389}$	$F(4, 9) = 3.63$
Within samples (Error)	3.50	$17 - 2 - 2$ $- 4 = 9$	$\frac{3.50}{9}$ $= 0.389$		
Total	76.28	$(18 - 1) = 17$			

Here,

d.f. for interaction = d.f. for columns $\times$ d.f. for rows.

d.f. for error = d.f. for total

– d.f. for column

– d.f. for rows

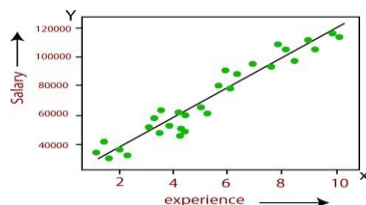
– d.f. for interactions

The above table shows that all the three F -ratios are significant at 5% level which means that the drugs act differently, different groups of people are affected differently and the interaction term is significant.

UNIT- 4

Linear Regression model :

- Linear regression is a statistical regression method which is used for predictive analysis.
- Linear regression shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis), hence called linear regression.
- If there is only one input variable (x), then such linear regression is called simple linear regression.
- The relationship between variables in the linear regression model can be explained using the below image. Here we are predicting the salary of an employee on the basis of the year of experience.



To calculate best-fit line linear regression uses a traditional slope intercept form which is given below,

$$Y_i = b_0 + b_1 * X_i$$

Where,

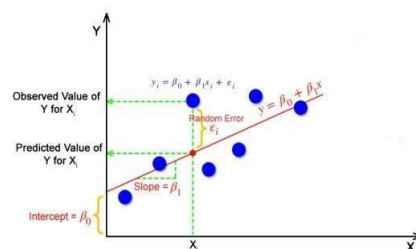
Y_i = Dependent variable

b_0 = constant/Intercept

b_1 = Slope

X_i = Independent variable.

- This algorithm explains the linear relationship between the dependent(output) variable y and the independent(predictor) variable X using a straight line $Y = B_0 + B_1 X$
- The goal of the linear regression algorithm is to get the best values for B_0 and B_1 to find the best-fit line. The best-fit line is a line that has the least error which means the error between predicted values and actual values should be minimum.



The least squares estimates of β_0 and β_1 are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Example 1:

The following data represents the no. of hours of 12 different students who watches the television & score of each student who has given test on Monday.

a) Find equation of regression line

b) Use the equation to find expected score of students who watches 9 hrs of television.

Hours(X)	0	1	2	3	3	5	5	5	6	7	7	10
Score(Y)	96	85	82	74	95	68	76	84	58	65	75	50

Ans:

Hours (x)	Score (y)	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
0	96	-4.5	20.334	20.25	-91.503
1	85	-3.5	9.334	12.25	-32.669
2	82	-2.5	6.334	6.25	-15.835
3	74	-1.5	-1.666	2.25	2.499
3	95	-1.5	19.334	2.25	-29.001
5	68	0.5	-7.666	0.25	-3.833
5	76	0.5	0.334	0.25	0.167
5	84	0.5	8.334	0.25	4.167
6	58	1.5	-17.666	2.25	-26.499
7	65	2.5	-10.666	6.25	-26.665
7	75	2.5	-0.666	6.25	-1.665
10	50	5.5	-25.666	30.25	-141.163
$\bar{x} = 4.5$	$\bar{y} = 75.666$			89	-362

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$= \frac{-362}{89} \quad = 75.666 - (-4.06)(4.5)$$

$$\hat{\beta}_1 = -4.06 \quad \hat{\beta}_0 = 57.396$$

a) $y_i = \beta_0 + \beta_1 x_i$

$$y_i = -4.06 + 93.936 x_i \quad y_i = 93.936 - 4.06(x_i)$$

b) $x_i = 9$

$$y_i = 93.936 - 4.06(9)$$

$$y_i = 57.396$$

Example 2:

Watching television also reduces the amount of physical exercise, causing weight gains. A sample of fifteen 10-year-old children was taken. The number of pounds each child was overweight was recorded (a negative number indicates the child is underweight). Additionally, the number of hours of television viewing per weeks was also recorded. These data are listed here: [OW: OverWeight]

TV	42	34	25	35	37	38	31	33	19	29	38	28	29	36	18
OW	18	6	0	-1	13	14	7	7	-9	8	8	5	3	14	-7

Fit the regression line and describe what the coefficients tell you about the relationship between the two variables.

TV (x)	Overweight (y)	$(x_i - \bar{x})$	$(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
42	18	10.534	12.5670	110.965	129.220
34	6	2.5340	0.2670	6.4212	0.676
25	0	-6.466	-5.7330	41.809	37.069
35	-1	3.5340	-6.7330	12.489	23.794
37	13	5.5340	7.2670	30.625	40.215
38	14	6.5340	8.2670	42.693	54.016
31	7	-0.466	1.2670	0.217	0.590
33	7	1.534	1.2670	2.3532	1.943
19	-9	-12.466	-14.7330	155.401	183.661
29	8	-2.466	2.2670	6.081	5.590
38	8	6.534	2.2670	42.693	14.8126
28	5	-3.466	-0.7330	12.013	2.5406
29	3	-2.466	-2.7330	6.081	6.7396
36	14	4.5340	8.2670	20.557	37.482
18	-7	-13.466	-12.7330	181.333	141.4626
$\bar{x} = 31.466$ $\bar{y} = 5.733$				671.731	649.8634

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_1 = \frac{649.8634}{671.731}$$

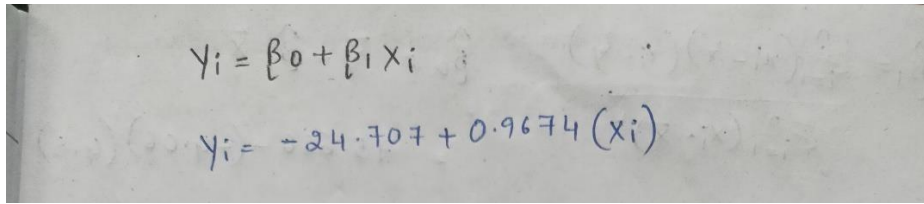
$$\hat{\beta}_1 = 0.9674$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\beta_0 = 5.733 - (0.9674)(31.466)$$

$$\beta_0 = 5.733 - 30.4402$$

$$\beta_0 = -24.707$$


$$y_i = \beta_0 + \beta_1 x_i$$
$$y_i = -24.707 + 0.9674(x_i)$$

Least square estimate:

The Least Squares Estimate (LSE) is a method used in statistical modeling to find the best-fitting line or curve by minimizing the sum of the squared differences between observed and predicted values. It is widely applied in regression analysis, where the goal is to estimate the relationship between a dependent variable (Y) and one or more independent variables (X).

Mathematically, in simple linear regression, the model is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where:

- Y = dependent variable (response)
- X = independent variable (predictor)
- β_0, β_1 = regression coefficients to be estimated
- ϵ = error term (random noise)

Importance of Least Squares Estimate in Regression Modeling

1. Best Fit Line Determination

- LSE finds the regression line that minimizes the overall prediction error, making it the most reliable method for estimating relationships.

2. Error Minimization

- Since squared errors are minimized, extreme values have less impact compared to other error measures, ensuring stable estimates.

3. Predictive Accuracy

- LSE improves the accuracy of predictive models by providing unbiased and efficient estimators of regression coefficients.

4. Foundation for Machine Learning Models

- Many machine learning models, such as **linear regression**, **ridge regression**, and **neural networks**, are based on minimizing loss functions similar to LSE.

5. Interpretability in Data Science

- Helps in understanding variable relationships, identifying trends, and making data-driven decisions in domains like **finance, healthcare, and crime analytics** (such as in your **Women Safety Analytics** project).

Multiple Linear Regression Model :

- In Simple Linear Regression, single Independent/Predictor(X) variable is used to model the response variable (Y). But there may be various cases in which the response variable is affected by more than one predictor variable; for such cases, the Multiple Linear Regression algorithm is used.
- Multiple Linear Regression is one of the important regression algorithms which models the linear relationship between a single dependent continuous variable and more than one independent variable.

Example:

Prediction of CO2 emission based on engine size and number of cylinders in a car.

- For MLR, the dependent or target variable(Y) must be the continuous/real, but the predictor or independent variable may be of continuous or categorical form.
- Each feature variable must model the linear relationship with the dependent variable.
- MLR tries to fit a regression line through a multidimensional space of data-points

MLR equation:

In Multiple Linear Regression, the target variable(Y) is a linear combination of multiple predictor variables $x_1, x_2, x_3, \dots, x_k$. Since it is an enhancement of Simple Linear Regression, so the same is applied for the multiple linear regression equation, the equation becomes:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \epsilon$$

The variable Y is dependent or predicted

The slope of y depends on the y-intercept

The regression coefficients β_1 and β_2 represent the change in y as a result of one-unit changes in x_1 and x_2

Assumptions for Multiple Linear Regression:

- A linear relationship should exist between the Target and predictor variables.
- The regression residuals must be normally distributed.
- MLR assumes little or no multicollinearity (correlation between the independent variable) in data.

Example Scenario:

Suppose we want to predict the sales (Y) of a product based on advertising spending on TV

(X1) radio (X2), and newspaper (X3). We have data on these variables for a number of

observations.

□ **Predictors (Independent Variables):**

- o X1: Advertising expenditure on TV.
- o X2: Advertising expenditure on radio.
- o X3: Advertising expenditure on newspaper.

□ **Response (Dependent Variable):**

- o Y: Sales of the product.

We have the following data:

TV Ad Spend (X1)	Radio Ad Spend (X2)	Newspaper Ad Spend (X3)	Sales (Y)
230	37	69	22
44	39	45	10
17	45	69	9
185	41	58	18

Steps to Build the Multiple Regression Model:

1. Data Exploration:
2. Fit the Regression Model:
3. Model Assessment
4. Interpretation of Coefficients

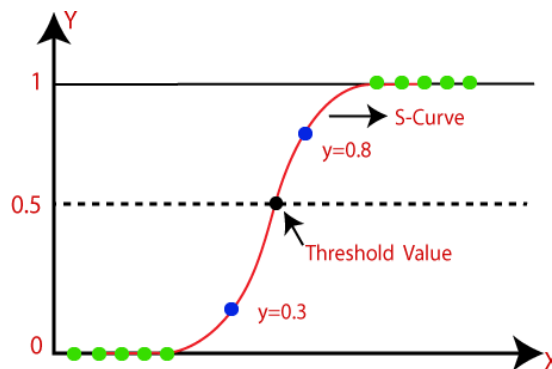
Summary:

Multiple Regression is a powerful statistical tool for analyzing relationships between multiple variables. It allows us to understand how changes in one or more predictors are associated with changes in the response variable, and helps in making predictions and understanding the relative importance of different predictors. It is widely used in fields such as economics, marketing, social sciences, and biomedical research to explore and quantify complex relationships in data.

Logistic Regression model:

- It is used for predicting the categorical dependent variable using a given set of independent variables.
- Logistic regression predicts the output of a categorical dependent variable. Therefore, the outcome must be a categorical or discrete value. It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, it gives the probabilistic values which lie between 0 and 1.

- Logistic Regression can be used to classify the observations using different types of data and can easily determine the most effective variables used for the classification. The below image is showing the logistic function:

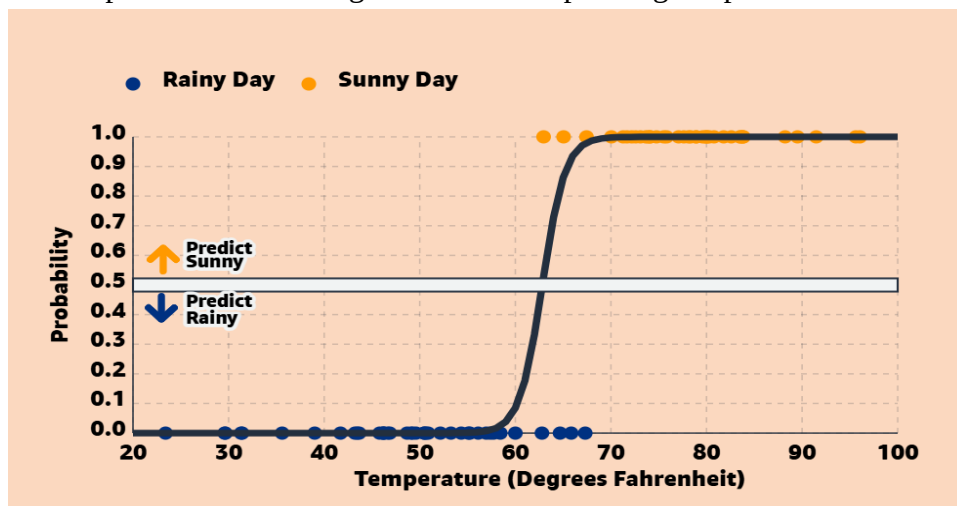


Types of Logistic Regression

- On the basis of the categories, Logistic Regression can be classified into three types:
 - Binomial:** In binomial Logistic regression, there can be only two possible types of the dependent variables, such as 0 or 1, Pass or Fail, etc.
 - Multinomial:** In multinomial Logistic regression, there can be 3 or more possible **unordered types** of the dependent variable, such as “cat”, “dogs”, or “sheep”
 - Ordinal:** In ordinal Logistic regression, there can be 3 or more possible **ordered types** of dependent variables, such as “low”, “Medium”, or “High”.

Example:

- You want to predict whether it will be sunny or rainy, so that you can decide whether to hike or not.
- Assume there are two classes: Rainy Days and Sunny Days. We can assign a numeric value of 0 and 1 to each class, say 0 to a Rainy Day and 1 to a Sunny Day.
- We have one continuous feature: the temperature, in degrees Fahrenheit. For each day, we can plot this value along with the corresponding temperature



- Clearly, we should not fit a linear regression model to these data. The outcomes of a linear regression model can take any numerical value, but these data can only take on outcomes of 0 or 1, so the predictions of a linear model may not be meaningful.

- Instead, we can fit a logistic function to the data. The values of this function can be interpreted as probabilities, as the values range between 0 and 1. We can interpret the line as the probability of a sunny day given a particular temperature.
- A typical threshold is 0.5, where we predict an outcome will occur if the probability of that outcome is greater than 0.5.

Prediction

- Temperature: 33 Degrees Fahrenheit : Rainy Day
- Temperature: 66 Degrees Fahrenheit : Sunny Day

Regression Analysis:

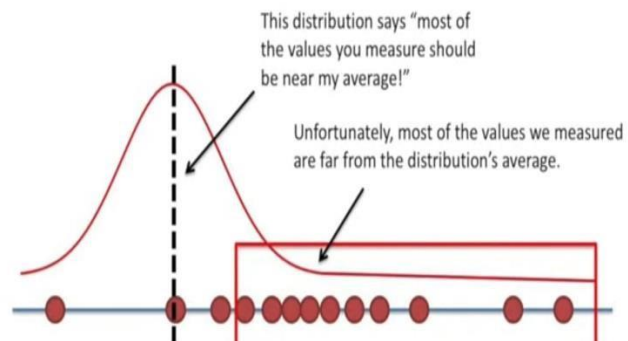
- Regression analysis is a statistical method to model the relationship between a dependent (target) and independent (predictor) variables with one or more independent variables.
- Regression analysis helps us to understand how the value of the dependent variable is changing corresponding to an independent variable when other independent variables are held fixed. It predicts continuous/real values such as temperature, age, salary, price, etc.

Some examples of regression can be as:

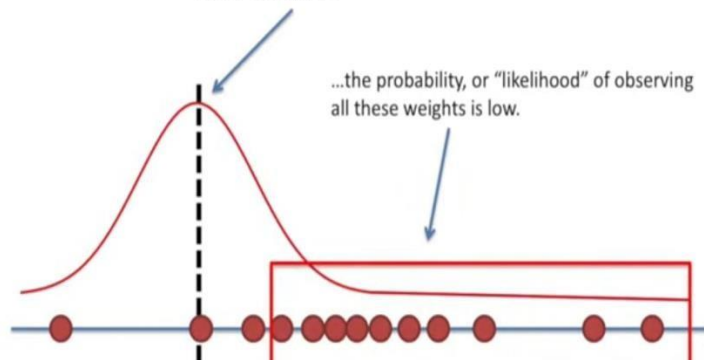
- Prediction of rain using temperature and other factors
- Determining Market trends
- Prediction of road accidents due to rash driving.

Maximum Likelihood Estimation (MLE)

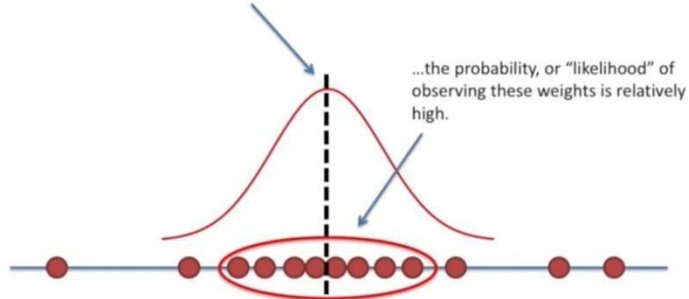
- Maximum Likelihood Estimation (MLE) is a method used to estimate the parameters of logistic regression by **maximizing the likelihood function**, which represents the probability of observing the given dataset.
- It works by calculating the likelihood, which measures how well the model explains the data for a given set of parameters. MLE seeks the parameter values that maximize this likelihood, effectively identifying the most plausible estimates based on the data.
- For example, if flipping a coin results in 7 heads out of 10 flips, MLE would estimate the probability of heads as 0.7, as this value best explains the observed outcome.
- Estimators are tools that use your data to estimate the values of parameters you're interested in. Common examples include the sample mean (used to estimate the average of a distribution) and the sample variance (used to estimate the spread of a distribution).



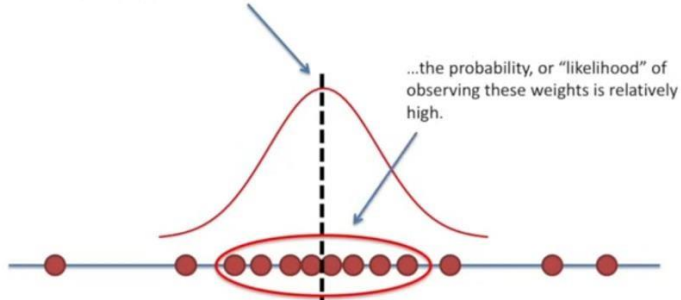
According to a normal distribution with a mean value over here...

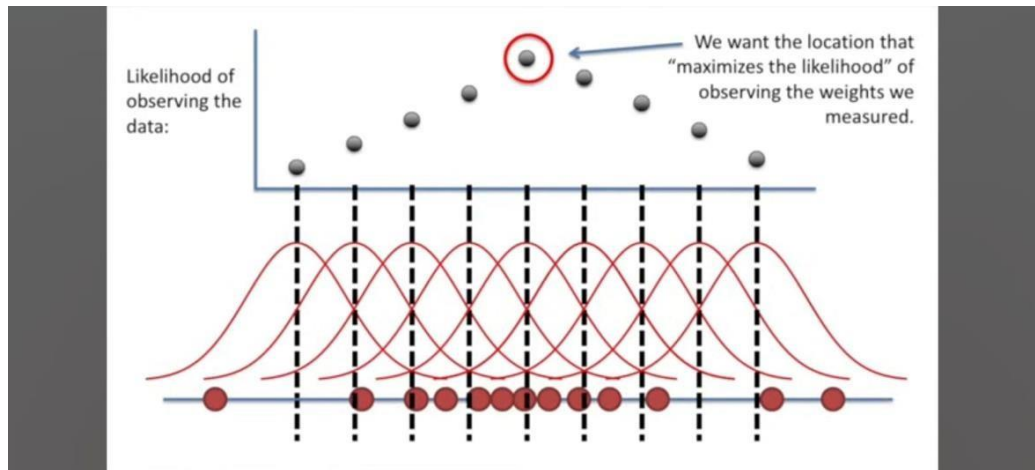


According to a normal distribution with a mean value here...



According to a normal distribution with a mean value here...





Linear Regression and Logistic Regression models:

Similarities

Aspect	Explanation
Supervised Learning	Both are supervised learning algorithms used to model relationships between dependent and independent variables.
Predictor Variables	Both can handle one or multiple independent (predictor) variables — either continuous or categorical (after encoding).
Linearity in Parameters	Both models assume a linear relationship between the predictors and the transformed output (log-odds in logistic regression).
Optimization	Both models typically use optimization algorithms (like gradient descent) to minimize a loss function.

Differences

Aspect	Linear Regression	Logistic Regression
Purpose	Predicts continuous numeric values.	Predicts categorical outcomes (often binary: 0 or 1).
Output	A continuous number (e.g., price, temperature).	A probability between 0 and 1, later classified into categories.
Equation	$y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n + \epsilon$	$P(y=1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}}$
Linearity	Models a straight-line relationship.	Models the log-odds of the dependent variable as a linear combination of predictors.
Loss Function	Uses Mean Squared Error (MSE).	Uses Log Loss (Cross-Entropy Loss).
Aspect	Linear Regression	Logistic Regression
Output Range	Output can be any real number from $-\infty$ to $+\infty$.	Output is bounded between 0 and 1.

Example	Predicting house prices based on size, number of rooms, location.	Predicting whether an email is spam (1) or not spam (0) based on its features.
Evaluation Metrics	R ² score, MSE, RMSE, MAE.	Accuracy, Precision, Recall, F1 Score, ROC-AUC.
Interpretation of Coefficients	Change in outcome per unit change in predictor.	Change in log-odds per unit change in predictor.

Example Illustration:

1. Linear Regression Example:

Problem: Predicting salary based on years of experience.

Model Output: "With 5 years of experience, the predicted salary is \$60,000."

2. Logistic Regression Example:

Problem: Predicting if a customer will buy a product based on age and income.

Model Output: "There is a 78% probability the customer will purchase; classified as YES."

Types of evaluation metrics for regression model:

Evaluation Metrics for Regression Model

Some common regression metrics

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R-squared (R²) Score
- Root Mean Squared Error (RMSE)

1. Mean Absolute Error (MAE): It's a measurement of the typical absolute discrepancies between a dataset's actual values and projected values.

Mathematical Formula

- The formula to calculate MAE for a data with "n" data points is:

$$MAE = \frac{1}{n} \sum_{i=1}^n |x_i - y_i|$$

Where:

- x_i represents the actual or observed values for the i-th data point.
- y_i represents the predicted value for the i-th data point

2. Mean Squared Error (MSE)

- It measures the square root of the average discrepancies between a dataset's actual values and projected values.
- MSE is frequently utilized in regression issues and is used to assess how well predictive models work.

Mathematical Formula

- For a dataset containing 'n' data points, the MSE calculation formula is:

$$MSE = \frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2$$

3. R-squared (R²) Score

- A statistical metric frequently used to assess the goodness of fit of a regression model is the **R-squared (R^2)** score, also referred to as the **coefficient of determination**.
- It quantifies the percentage of the dependent variable's variation that the model's independent variables contribute to.
- R^2 is a useful statistic for evaluating the overall effectiveness and explanatory power of a regression model.

Mathematical Formula

- The formula to calculate the R-squared score is as follows:

$$R^2 = 1 - \frac{SSR}{SST}$$

Where:

- R^2 is the R-Squared.
- SSR represents the sum of squared residuals between the predicted values and actual values.
- SST represents the total sum of squares, which measures the total variance in the dependent variable.

4. Root Mean Squared Error (RMSE)

- RMSE stands for Root Mean Squared Error.
- It is a usually used metric in regression analysis and machine learning to measure the accuracy or goodness of fit of a predictive model, especially when the predictions are continuous numerical values

Mathematical Formula

The formula for RMSE for a data with 'n' data points is as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2}$$

Where:

- RMSE is the Root Mean Squared Error.
- x_i represents the actual or observed value for the i-th data point.
- y_i represents the predicted value for the i-th data point.

Principal Component Analysis

- Having too many features in data can cause problems like overfitting (good on training data but poor on new data), slower computation, and lower accuracy.
- The explosion of feature combinations makes sampling harder In high-dimensional data and tasks like clustering or classification more complex and slow.
- To tackle this problem, we use Feature engineering Techniques, such as feature selection and feature extraction. One popular feature extraction method is dimensionality reduction, which reduces the number of features while keeping as much important information as possible.

- One of the most widely used dimensionality reduction techniques is Principal Component Analysis (PCA).

How PCA Works for Dimensionality Reduction?

• Step 1: Standardize the Data

Make sure all features (e.g., height, weight, age) are on the same scale. Why? A feature like “salary” (ranging 0–100,000) could dominate “age” (0–100) otherwise.

Standardizing our dataset to ensure that each variable has a mean of 0 and a standard deviation of 1.

$$Z = \frac{X - \mu}{\sigma}$$

Here,

- μ is the mean of independent features $\mu = \{\mu_1, \mu_2, \dots, \mu_m\}$
- σ is the standard deviation of independent features $\sigma = \{\sigma_1, \sigma_2, \dots, \sigma_m\}$

• Step 2: Find Relationships

Calculate how features **move together** using a *covariance matrix*. Covariance measures the strength of joint variability between two or more variables, indicating how much they change in relation to each other. To find the covariance we can use the formula:

$$\text{cov}(x_1, x_2) = \frac{\sum_{i=1}^n (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)}{n-1}$$

The value of covariance can be positive, negative, or zeros.

- **Positive:** As the x_1 increases x_2 also increases.
- **Negative:** As the x_1 increases x_2 also decreases.
- **Zeros:** No direct relation.

Step 3: Find the “Magic Directions” (Principal Components)

- PCA identifies **new axes** (like rotating a camera) where the data spreads out the most:
 - **1st Principal Component (PC1):** The direction of maximum variance (most spread).
 - **2nd Principal Component (PC2):** The next best direction, *perpendicular to PC1*, and so on.

Step 4: Pick the Top Directions & Transform Data

- Keep only the top 2–3 directions (or enough to capture ~95% of the variance).
- Project the data onto these directions to get a simplified, lower-dimensional version.

Advantages of Principal Component Analysis

- **Multicollinearity Handling:** Creates new, uncorrelated variables to address issues when original features are highly correlated.
- **Noise Reduction:** Eliminates components with **low variance** (assumed to be noise), enhancing data clarity.

- **Data Compression:** Represents **data with fewer components**, reducing storage needs and speeding up processing.
- **Outlier Detection:** Identifies unusual data points by showing which ones deviate significantly in the reduced space.

Disadvantages of Principal Component Analysis

- **Interpretation Challenges:** The new components are combinations of original variables, which can be hard to explain.
- **Data Scaling Sensitivity:** Requires proper scaling of data before application, or results may be misleading.
- **Information Loss:** Reducing dimensions may lose some important information if too few components are kept.
- **Assumption of Linearity:** Works best when relationships between variables are linear, and may struggle with non-linear data.
- **Computational Complexity:** Can be slow and resource-intensive on very large datasets.
- **Risk of Overfitting:** Using too many components or working with a small dataset might lead to models that don't generalize well.

UNIT- 5
Predictive Analysis

Supervised Learning Algorithm

- **Supervised Machine Learning** is a fundamental approach for machine learning and artificial intelligence.
- It involves training a model using labeled data, where each input comes with a corresponding correct output.
- The process is like a teacher guiding a student—hence the term “supervised” learning.
- In this article, we’ll explore the key components of supervised learning, the different types of supervised machine learning algorithms used, and some practical examples of how it works.

INPUT RAW DATA

Labeled Data



Labels



Supervised Machine Learning

Model is trained using labeled data and pre-processed to predict outcomes and classify data accurately.

OUTPUT

Processing

Algorithm

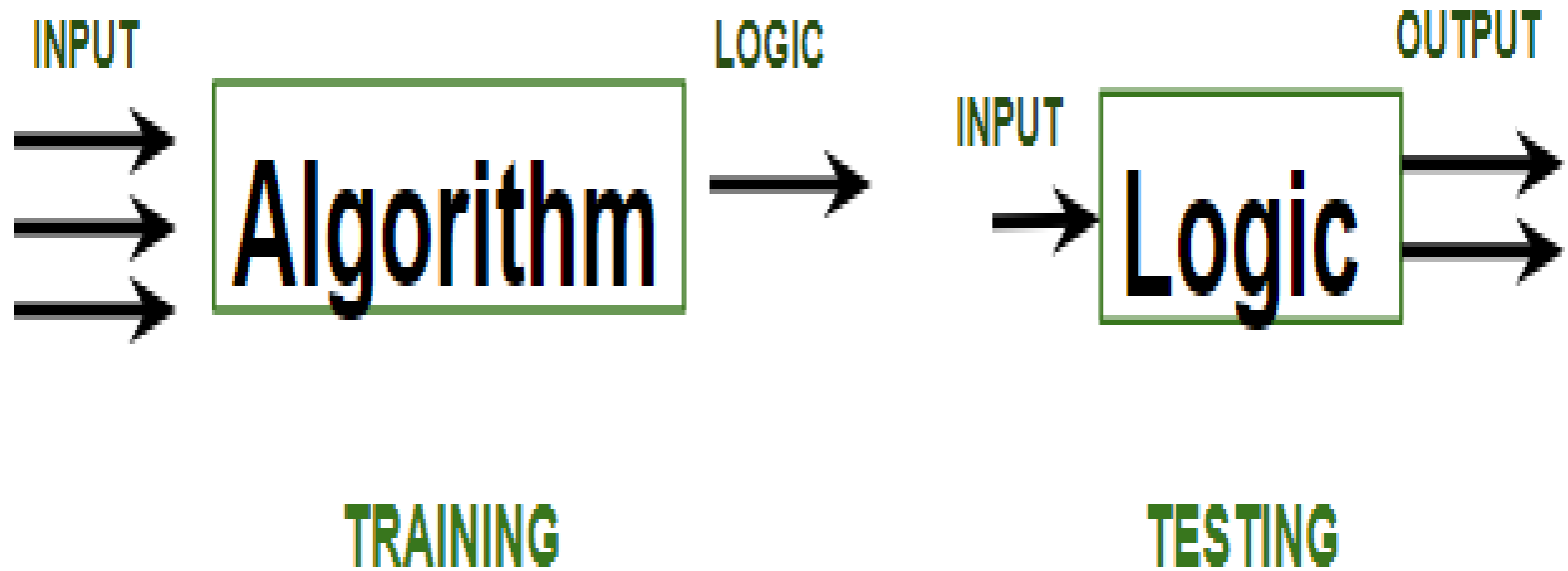


Elephant

Camel

Cow

- **supervised learning** is a type of machine learning where a model is trained on labeled data—meaning each input is paired with the correct output. the model learns by comparing its predictions with the actual answers provided in the training data. Over time, it adjusts itself to minimize errors and improve accuracy.
- The goal of supervised learning is to make accurate predictions when given new, unseen data.
- For example, if a model is trained to recognize handwritten digits, it will use what it learned to correctly identify new numbers it hasn't seen before.
- Supervised learning can be applied in various forms, including supervised learning classification and supervised learning regression, making it a crucial technique in the field of artificial intelligence and supervised data mining.
- Where **Supervised Learning Algorithm** consists of input features and corresponding output labels. The process works through:
 - **Training Data:** The model is provided with a training dataset that includes input data (features) and corresponding output data (labels or target variables).
 - **Learning Process:** The algorithm processes the training data, learning the relationships between the input features and the output labels.
 - This is achieved by adjusting the model's parameters to minimize the difference between its predictions and the actual labels.
 - **Training** phase involves feeding the algorithm labeled data, where each data point is paired with its correct output. The algorithm learns to identify patterns and relationships between the input and output data.
 - **Testing** phase involves feeding the algorithm new, unseen data and evaluating its ability to predict the correct output based on the learned patterns.



Practical Examples of Supervised learning

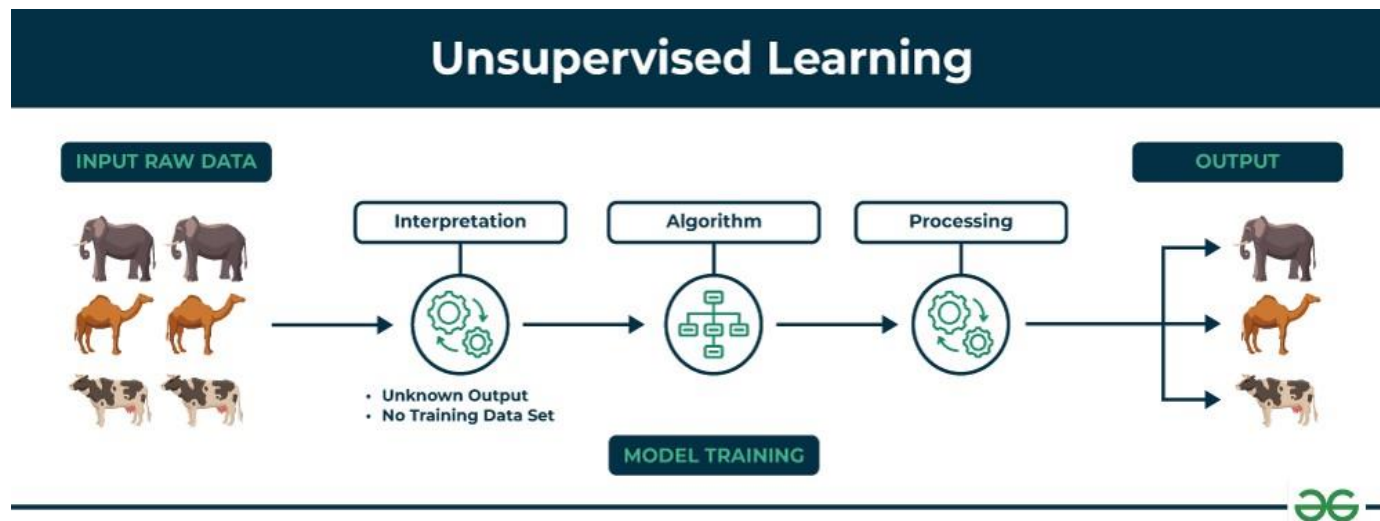
- Few practical examples of supervised machine learning across various industries:
- **Fraud Detection in Banking:** Utilizes supervised learning algorithms on historical transaction data, training models with labeled datasets of legitimate and fraudulent transactions to accurately predict fraud patterns.
- **Parkinson Disease Prediction:** Parkinson's disease is a progressive disorder that affects the nervous system and the parts of the body controlled by the nerves.
- **Customer Churn Prediction:** Uses supervised learning techniques to analyze historical customer data, identifying features associated with churn rates to predict customer retention effectively.
- **Cancer cell classification:** Implements supervised learning for cancer cells based on their features, and identifying them if they are
'malignant' or 'benign'.
- **Stock Price Prediction:** Applies supervised learning to predict a signal that indicates whether buying a particular stock will be helpful or not.

Supervised Machine Learning Algorithms

- **Supervised learning** can be further divided into several different types, each with its own unique characteristics and applications. Here are some of the most common types of supervised learning algorithms:
- **Linear Regression**: Linear regression is a type of supervised learning regression algorithm that is used to predict a continuous output value. It is one of the simplest and most widely used algorithms in supervised learning.
- **Logistic Regression**: Logistic regression is a type of supervised learning classification algorithm that is used to predict a binary output variable.
- **Decision Trees**: Decision tree is a tree-like structure that is used to model decisions and their possible consequences. Each internal node in the tree represents a decision, while each leaf node represents a possible outcome.
- **Random Forests**: Random forests again are made up of multiple decision trees that work together to make predictions. Each tree in the forest is trained on a different subset of the input features and data. The final prediction is made by aggregating the predictions of all the trees in the forest.
- **Support Vector Machine(SVM)**: The SVM algorithm creates a hyperplane to segregate n-dimensional space into classes and identify the correct category of new data points. The extreme cases that help create the hyperplane are called support vectors, hence the name Support Vector Machine.
- **K-Nearest Neighbors (KNN)**: KNN works by finding k training examples closest to a given input and then predicts the class or value based on the majority class or average value of these neighbors. The performance of KNN can be influenced by the choice of k and the distance metric used to measure proximity.
- **Gradient Boosting**: Gradient Boosting combines weak learners, like [decision trees](#), to create a strong model. It iteratively builds new models that correct errors made by previous ones.
- **Naive Bayes Algorithm**: The Naive Bayes algorithm is a supervised machine learning algorithm based on applying [Bayes' Theorem](#) with the “naive” assumption that features are independent of each other given the class label.

Unsupervised Learning Algorithm

- Unsupervised learning algorithms are tasked with finding patterns and relationships within the data without any prior knowledge of the data's meaning.
- Unsupervised machine learning algorithms find hidden patterns and data without any human intervention, i.e., we don't give output to our model.
- The training model has only input parameter values and discovers the groups or patterns on its own.



- **The image shows set of animals:** elephants, camels, and cows that represents raw data that the unsupervised learning algorithm will process.
- **Interpretation** stage signifies that the algorithm doesn't have predefined labels or categories for the data. It needs to figure out how to group or organize the data based on inherent patterns.
- **Algorithm** represents the core of unsupervised learning process using techniques like clustering, dimensionality reduction, or anomaly detection to identify patterns and structures in the data.
- **Processing** stage shows the algorithm working on the data.
- **How it works:** Unsupervised learning works by analyzing unlabeled data to identify patterns and relationships. The data is not labeled with any predefined categories or outcomes, so the algorithm must find these patterns and relationships on its own.
- This can be a challenging task, but it can also be very rewarding, as it can reveal insights into the data that would not be apparent from a labeled dataset.

Challenges of Unsupervised Learning:

Here are the key challenges of unsupervised learning:

Noisy Data: Outliers and noise can distort patterns and reduce the effectiveness of algorithms.

Assumption Dependence: Algorithms often rely on assumptions (e.g., cluster shapes), which may not match the actual data structure.

Overfitting Risk: Overfitting can occur when models capture noise instead of meaningful patterns in the data.

Limited Guidance: The absence of labels restricts the ability to guide the algorithm toward specific outcomes.

Cluster Interpretability: Results, such as clusters, may lack clear meaning or alignment with real-world categories.

Sensitivity to Parameters: Many algorithms require careful tuning of hyperparameters, such as the number of clusters in k-means.

Lack of Ground Truth: Unsupervised learning lacks labeled data, making it difficult to evaluate the accuracy of results.

Applications of Unsupervised learning:

Unsupervised learning has diverse applications across industries and domains.

Key applications include:

Customer Segmentation: Algorithms cluster customers based on purchasing behavior or demographics, enabling targeted marketing strategies.

Anomaly Detection: Identifies unusual patterns in data, aiding fraud detection, cybersecurity, and equipment failure prevention.

Recommendation Systems: Suggests products, movies, or music by analyzing user behavior and preferences.

Image and Text Clustering: Groups similar images or documents for tasks like organization, classification, or content recommendation.

Social Network Analysis: Detects communities or trends in user interactions on social media platforms.

Astronomy and Climate Science: Classifies galaxies or groups weather patterns to support scientific research

Model evaluation and performance metrics:

- Performance of diagnostic tests,
- confusion Matrix
- true and false positives, precision and recall measures.
- Roc curves,
- Area under the Curve,

Common Diagnostic Test Performance Indices

		Disease		
		Present	Absent	
Test Result	Positive	<i>TP</i>	<i>FP</i>	<i>TP + FP</i>
	Negative	<i>FN</i>	<i>TN</i>	<i>FN + TN</i>
		<i>TP + FN</i>	<i>FP + TN</i>	

TP: True-positive; *FP*: False-positive

FN: False-negative; *TN*: True-negative

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Positive Predictive Value} = \frac{TP}{TP + FP}$$

$$\text{Negative Predictive Value} = \frac{TN}{TN + FN}$$

$$\text{Positive Likelihood Ratio} = \frac{\text{Sensitivity}}{1 - \text{Specificity}}$$

$$\text{Negative Likelihood Ratio} = \frac{1 - \text{Sensitivity}}{\text{Specificity}}$$

The Confusion Matrix Structure:

Let's learn about the basic structure of a confusion matrix, using the example of identifying an email as spam or not spam.

True Positive (TP) - Your model predicted the positive class. For example, identifying a spam email as spam.

True Negative (TN) - Your model correctly predicted the negative class. For example, identifying a regular email as not spam.

False Positive (FP) - Your model incorrectly predicted the positive class. For example, identifying a regular email as spam.

False Negative (FN) - Your model incorrectly predicted the negative class. For example, identifying a spam email as a regular email.

		True Class	
		Positive	Negative
Predicted Class	Positive	TP	FP
	Negative	FN	TN

	Positive Predicted Negative	
Negative Actual Positive	True Positive (TP) Correct objects	False Negative (FN) Missed objects
	False Positive (FP) Extra objects	True Negative (TN) No objects
	Precision $\frac{TP}{(TP+FP)}$	Negative predictive value $\frac{TN}{(TN+TP)}$

$$\text{Recall} = \frac{TP}{(TP+FN)}$$

$$\text{Accuracy} = \frac{(TP+TN)}{(TP+TN+FP+FN)}$$

$$\text{Specificity} = \frac{TN}{(TN+FP)}$$

$$F1 = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}}$$

Calculating a Confusion Matrix

- Now that we have a good understanding of a basic confusion matrix, its terminology, and its use, let's move on to manually calculating a confusion matrix, followed by a practical example.

Manually calculating a confusion matrix

- Here is a step-by-step guide on how to manually calculate a Confusion Matrix.

1. Define the outcomes

- The first step will be to identify the two possible outcomes of your task: Positive or Negative.

2. Collecting the predictions

- Once your possible outcomes are defined, the next step will be to collect all the model's predictions, including how many times the model predicted each class and its occurrence.

3. Classifying the outcomes

- Once all the predictions have been collated, the next step is to classify the outcomes into the four categories:
 - **True Positive (TP)**
 - **True Negative (TN)**
 - **False Positive (FP)**
 - **False Negative (FN)**

- [Classification Accuracy](#)
- [Logarithmic loss](#)
- [Area under Curve](#)
- [F1 score](#)
 - Precision
 - Recall
- [Confusion Matrix](#)

Classification accuracy: is a fundamental metric for evaluating the performance of a classification model, providing a quick snapshot of how well the model is performing in terms of correct predictions. This is calculated as the ratio of correct predictions to the total number of input Samples.

- **Accuracy**=No. of correct predictions/Total number of input samples
- It works great if there are an equal number of samples for each class.
- **For example:** we have a 90% sample of *class A* and a 10% sample of *class B* in our training set. Then, our model will predict with an accuracy of 90% by predicting all the training samples belonging to *class A*. If we test the same model with a test set of 60% from class A and 40% from class B. Then the accuracy will fall, and we will get an accuracy of 60%.
- *In a Covid test of 1000 patients, there were 45 positive tests, of which 30 patients had covid and 15 were falsely tested positive. Of the 955 negative tests there were 5 that were incorrect, these patients had covid but were tested negatively.*
- *Draw the confusion matrix and calculate the accuracy, precision, recall, sensitivity and F1 score from the matrix.*

Logarithmic Loss: Log loss quantifies how well a model's predicted probabilities align with the true outcomes. It's a logarithmic transformation of the likelihood function, making it sensitive to both the correctness and confidence of predictions.

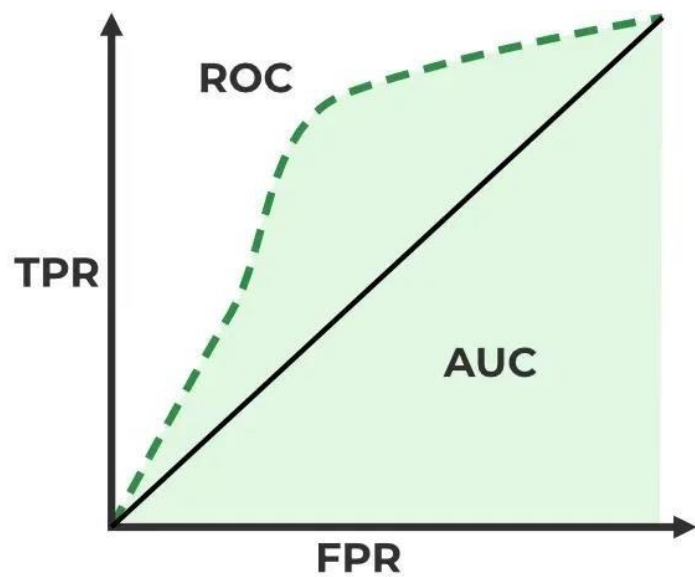
$$Logloss = \frac{1}{N} \sum_{i=1}^N logloss_i$$

$$Logloss = -\frac{1}{N} \sum_{i=1}^N [y_i \ln p_i + (1 - y_i) \ln(1 - p_i)]$$

- How it works: Log loss penalizes models for being overconfident in incorrect predictions or under confident in correct ones.
- It calculates the negative average of the logarithm of the predicted probabilities for the true class.
- The lower the log loss, the better the model's performance.

AUC-ROC Curve

- The **AUC-ROC curve** is an essential tool used for evaluating the performance of binary classification models. It plots the **True Positive Rate (TPR)** against the **False Positive Rate (FPR)** at different thresholds showing how well a model can distinguish between two classes such as positive and negative outcomes.
- It provides a graphical representation of the model's ability to distinguish between two classes like positive class for presence of a disease and negative class for absence of a disease.
- Key Terms in AUC-ROC:
- **TPR (True Positive Rate)**: The ratio of correctly predicted positive instances.
- **FPR (False Positive Rate)**: The ratio of incorrectly predicted negative instances.
- **Specificity**: The proportion of actual negatives correctly identified by the model (inverse of FPR).
- **Sensitivity/Recall**: The proportion of actual positives correctly identified by the model (same as TPR).



		Actual	
		Positive	Negative
Predicted	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

Understanding ROC and AUC:

- **ROC Curve:** [ROC Curve](#) plots TPR vs. FPR at different thresholds. It represents the trade-off between the sensitivity and specificity of a classifier.
- **AUC (Area Under the Curve):** [AUC](#) measures the area under the ROC curve. A higher AUC value indicates better model performance as it suggests a greater ability to distinguish between classes. An AUC value of 1.0 indicates perfect performance while 0.5 suggests it is random guessing.

How AUC-ROC Works

- AUC-ROC curve helps us understand how well a classification model distinguishes between the two classes (positive and negative).
- Imagine we have 6 data points and out of these:
- **3 belong to the positive class:** Class 1 for people who have a disease.
- **3 belong to the negative class:** Class 0 for people who don't have disease

AUC-ROC is effective when:

- The dataset is balanced and the model needs to be evaluated across all thresholds.
- False positives and false negatives are of similar importance.
- In cases of highly imbalanced datasets AUC-ROC might give overly optimistic results. In such cases the Precision-Recall Curve is more suitable focusing on the positive class.

Model Performance with AUC-ROC

- **High AUC (close to 1):** The model effectively distinguishes between positive and negative instances.
- **Low AUC (close to 0):** The model struggles to differentiate between the two classes.
- **AUC around 0.5:** The model doesn't learn any meaningful patterns i.e it is doing random guessing.
- In short, the **AUC** gives you an overall idea of how well your model is doing at sorting positives and negatives, without being affected by the threshold you set for classification. A higher **AUC** means your model is doing good.

Simple Machine Learning Applications

1. **Image Recognition:** Identifying objects, faces, and scenes in images.

- Machine learning algorithms can be trained to identify objects, faces, or scenes within images.

Examples:

- Facial recognition in smartphones, object detection in self-driving cars, and image tagging on social media.

2. **Speech Recognition:** Converting spoken words into text, powering voice assistants like Siri and Alexa.

How it works:

- Machine learning models can convert spoken words into text, enabling voice-controlled devices and applications.

Examples:

- Voice assistants like Siri and Alexa, dictation software, and speech-to-text applications.

3. Recommender Systems: Suggesting products, movies, or content based on user preferences.

How it works:

- Machine learning algorithms analyze user data (e.g., purchase history, browsing behavior) to suggest products they might be interested in.

Examples:

- E-commerce websites suggesting related products, streaming services recommending movies or shows, and social media platforms suggesting content or friends.

4. Spam Filtering: Identifying and filtering unwanted emails.

How it works:

- Machine learning models analyze email content and sender information to identify and filter out spam emails.

Examples:

- Email providers using machine learning to filter spam, and users using spam filters on their email clients.

5. Fraud Detection: Identifying suspicious financial transactions.

How it works:

- Machine learning models analyze financial transactions and user behavior to detect and prevent fraudulent activities.

Examples:

- Credit card companies using machine learning to flag suspicious transactions, and online banking platforms detecting potential fraud

6. Predictive Analytics: Forecasting future outcomes, such as sales or customer churn.

How it works:

- Machine learning models analyze customer data to predict which customers are likely to leave a company or service.

Examples:

- Telecommunication companies using machine learning to identify customers at risk of churn, and subscription services predicting customer cancellations.

7. Natural Language Processing (NLP): Enabling machines to understand and respond to human language.

How it works:

- Machine learning algorithms can be used to process and understand human language, enabling tasks like translation, sentiment analysis, and chatbot development.
- **Examples:**
- Google Translate, sentiment analysis tools for social media monitoring, and virtual assistants like Siri and Alexa.

Simple Machine Learning Algorithms:

- **Linear Regression:** Linear regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables. It provides valuable insights for prediction and data analysis.
- **Logistic Regression:** **Logistic regression** is a **supervised machine learning algorithm** used for **classification tasks** where the goal is to predict the probability that an instance belongs to a given class or not. Logistic regression is a statistical algorithm which analyze the relationship between two data factors. The article explores the fundamentals of logistic regression, it's types and implementations.
- Logistic regression is used for binary [classification](#) where we use [sigmoid function](#), that takes input as independent variables and produces a probability value between 0 and 1.
- **Decision Trees:** A decision tree is a graphical representation of different options for solving a problem and show how different factors are related. It has a hierarchical tree structure starts with one main question at the top called a node which further branches out into different possible outcomes where:

Root Node is the starting point that represents the entire dataset.

Branches: These are the lines that connect nodes. It shows the flow from one decision to another.

Internal Nodes are Points where decisions are made based on the input features.

Leaf Nodes: These are the terminal nodes at the end of branches that represent final outcomes or predictions
Using a tree-like structure to make decisions based on data features.

Unsupervised Learning

1. Some common clustering algorithms:

- **K-means Clustering**: Groups data into K clusters based on how close the points are to each other.
- **Hierarchical Clustering**: Creates clusters by building a tree step-by-step, either merging or splitting groups.
- **Density-Based Clustering (DBSCAN)**: Finds clusters in dense areas and treats scattered points as noise.
- **Mean-Shift Clustering**: Discovers clusters by moving points toward the most crowded areas.
- **Spectral Clustering**: Groups data by analyzing connections between points using graphs.

2. Dimensionality Reduction:

- Here are some popular **Dimensionality Reduction algorithms**:
- **Principal Component Analysis (PCA)**: Reduces dimensions by transforming data into uncorrelated principal components.
- **Linear Discriminant Analysis (LDA)**: Reduces dimensions while maximizing class separability for classification tasks.
- **Non-negative Matrix Factorization (NMF)**: Breaks data into non-negative parts to simplify representation.
- **Locally Linear Embedding (LLE)**: Reduces dimensions while preserving the relationships between nearby points.
- **Isomap**: Captures global data structure by preserving distances along a manifold.

3. Some common Association Rule Learning algorithms:

- **Apriori Algorithm**: Finds patterns by exploring frequent item combinations step-by-step.
- **FP-Growth Algorithm**: An Efficient Alternative to Apriori. It quickly identifies frequent patterns without generating candidate sets.
- **Eclat Algorithm**: Uses intersections of itemsets to efficiently find frequent patterns.
- **Efficient Tree-based Algorithms**: Scales to handle large datasets by organizing data in tree structures.